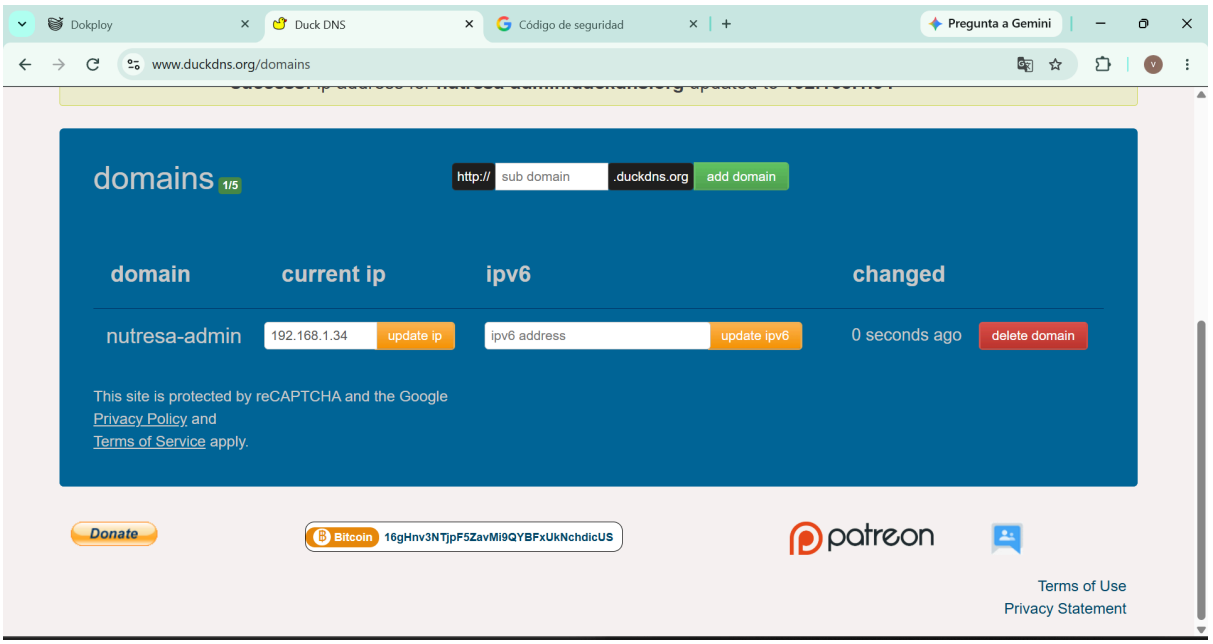
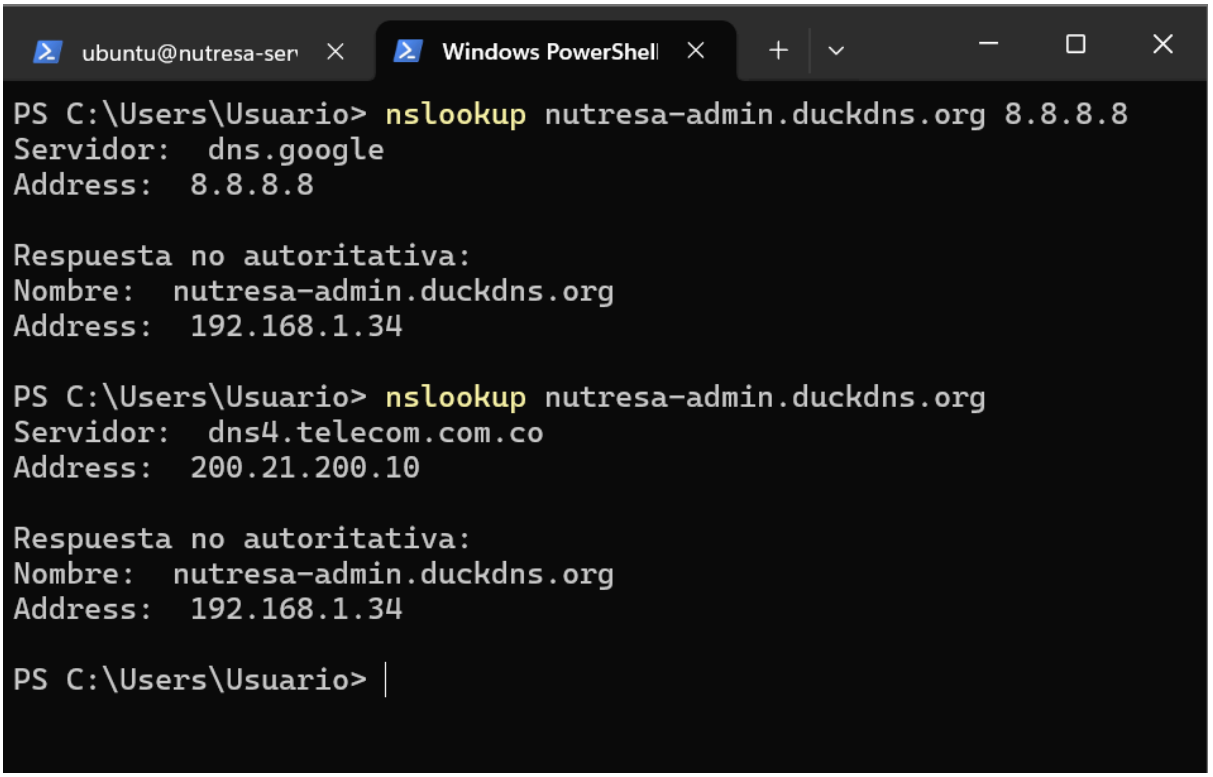
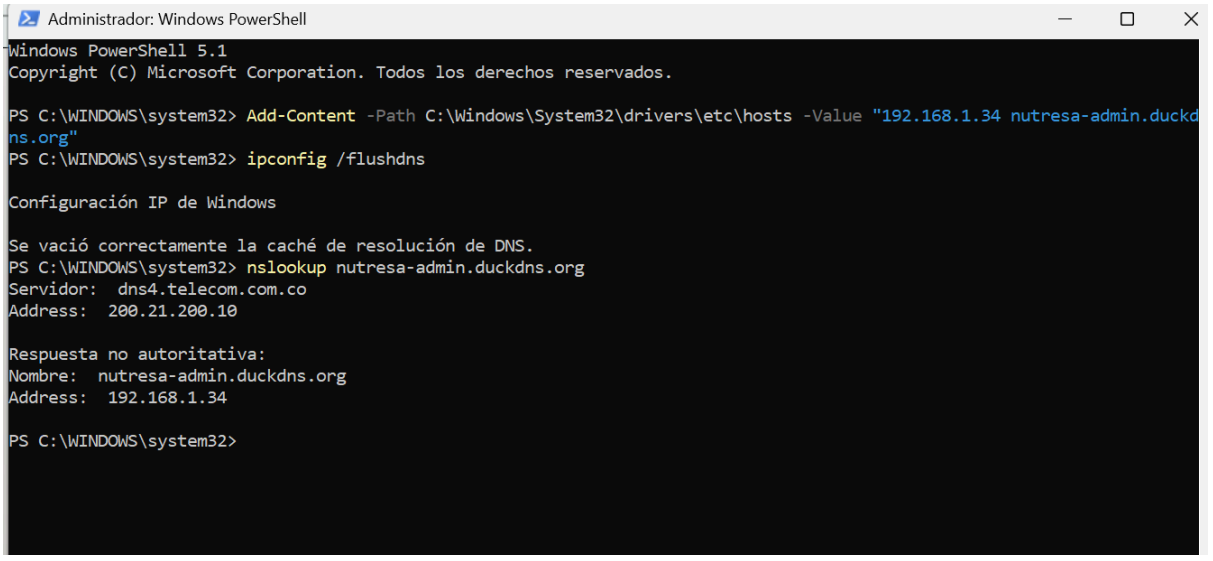
ADMIN TI CAPTURAS

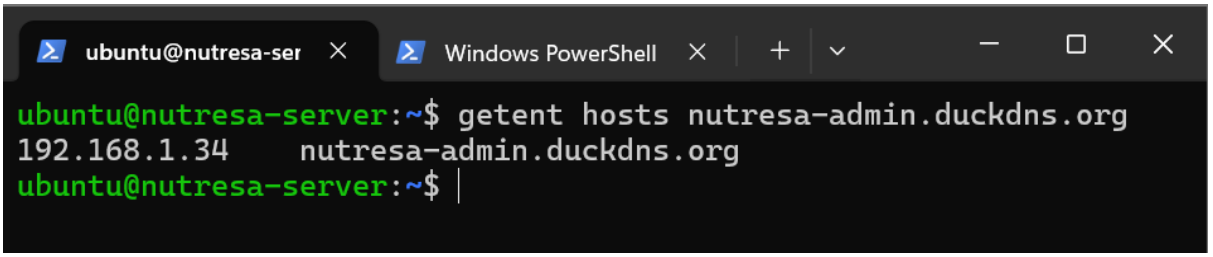
SEMANA 2
DNS Y SSL






ubuntu@nutresa-ser
Windows PowerShell
PS C:\Users\Usuario> nslookup nutresa-admin.duckdns.org 8.8.8.8
Servidor: dns.google
Address: 8.8.8.8
Respuesta no autoritativa:
Nombre: nutresa-admin.duckdns.org
Address: 192.168.1.34
PS C:\Users\Usuario> nslookup nutresa-admin.duckdns.org
Servidor: dns4.telecom.com.co
Address: 200.21.200.10
Respuesta no autoritativa:
Nombre: nutresa-admin.duckdns.org
Address: 192.168.1.34
PS C:\Users\Usuario>


Administrador: Windows PowerShell
Windows PowerShell 5.1
Copyright (C) Microsoft Corporation. Todos los derechos reservados.
PS C:\WINDOWS\system32> Add-Content -Path C:\Windows\System32\drivers\etc\hosts -Value "192.168.1.34 nutresa-admin.duckdns.org"
PS C:\WINDOWS\system32> ipconfig /flushdns
Configuración IP de Windows
Se vació correctamente la caché de resolución de DNS.
PS C:\WINDOWS\system32> nslookup nutresa-admin.duckdns.org
Servidor: dns4.telecom.com.co
Address: 200.21.200.10
Respuesta no autoritativa:
Nombre: nutresa-admin.duckdns.org
Address: 192.168.1.34
PS C:\WINDOWS\system32>


ubuntu@nutresa-ser
Windows PowerShell
ubuntu@nutresa-server:~$ getent hosts nutresa-admin.duckdns.org
192.168.1.34   nutresa-admin.duckdns.org
ubuntu@nutresa-server:~$

```
ubuntu@nutresa-server:~$ sudo docker inspect dokploy-traefik --f
ormat '{{range .Config.Env}}{{println .}}{{end}}'
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bi
n

ubuntu@nutresa-server:~$ sudo docker inspect dokploy-traefik --f
ormat '{{json .NetworkSettings.Networks}}' | grep -o '"dokploy-n
etwork"'
"dokploy-network"
ubuntu@nutresa-server:~$
```

```
ubuntu@nutresa-server:~$ sudo docker mkdir -p /root/backup-red
unknown shorthand flag: 'p' in -p

Usage:  docker [OPTIONS] COMMAND [ARG...]

Run 'docker --help' for more information
ubuntu@nutresa-server:~$ sudo mkdir -p /root/backup-red
ubuntu@nutresa-server:~$ sudo cp -a /etc/dokploy/traefik/traefik.yml /root/backup-red/traefik.yml.bak
ubuntu@nutresa-server:~$ sudo ls -l /root/backup-red
total 8
-rw------- 1 root root 330 Nov 10 15:46 50-cloud-init.yaml.bak
-rw-r--r-- 1 root root 612 Nov 10 22:49 traefik.yml.bak
ubuntu@nutresa-server:~$
```

```
ubuntu@nutresa-server:~$ read -rsp "b33a09a3-3774-4f69-ae3c-70581989b305
> " T; echo
b33a09a3-3774-4f69-ae3c-70581989b305

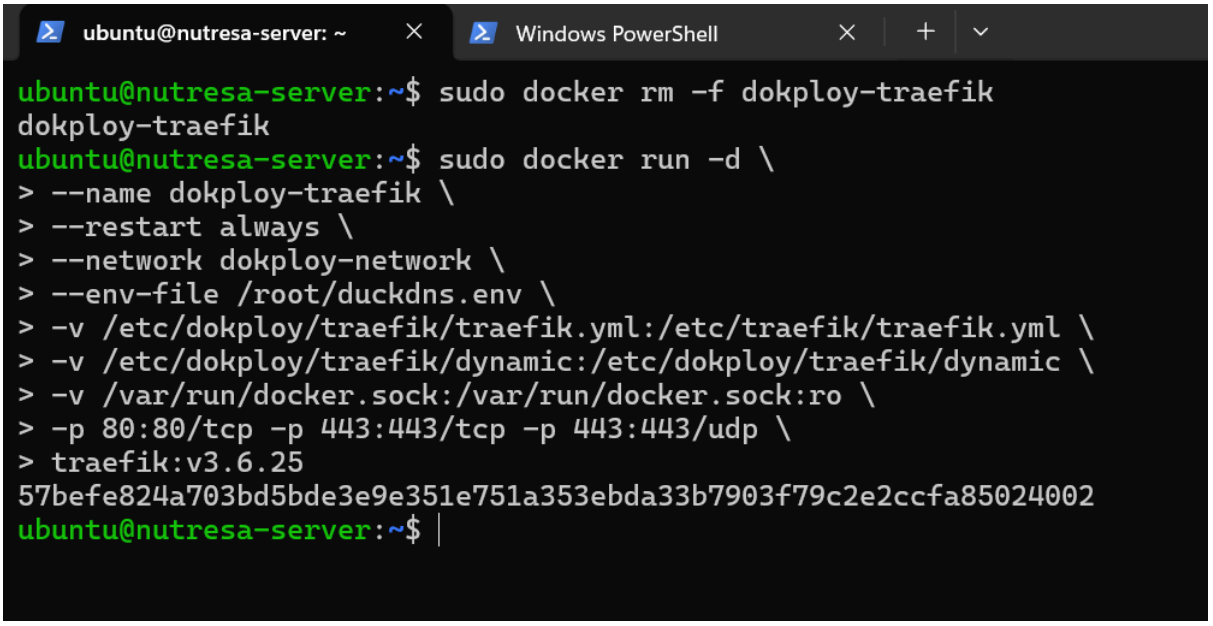
ubuntu@nutresa-server:~$ printf 'DUCKDNS_TOKEN=%s\n' "$T" | sudo tee /root/duckdns.env >/dev/null
ubuntu@nutresa-server:~$ sudo chmod 600 /root/duckdns.env
ubuntu@nutresa-server:~$ unset T
ubuntu@nutresa-server:~$ sudo ls -l /root/duckdns.env
-rw------- 1 root root 15 Nov 12 15:45 /root/duckdns.env
ubuntu@nutresa-server:~$
```

```
ubuntu@nutresa-server: ~   Windows PowerShell
  GNU nano 7.2                                  /etc/dokploy/traefik/traefik.yml *
  exposedByDefault: false
  watch: true
  network: dokploy-network
 file:
  directory: /etc/dokploy/traefik/dynamic
  watch: true
entryPoints:
 web:
  address: :80
 websecure:
  address: :443
  http3:
   advertisedPort: 443
  http:
   tls:
    certResolver: letsencrypt
api:
 insecure: true
certificatesResolvers:
 letsencrypt:
  acme:
   email: valentina.gutierrezc@uqvirtual.edu.co
   storage: /etc/dokploy/traefik/dynamic/acme.json
   caServer: https://acme-staging-v02.api.letsencrypt.org/directory
   dnsChallenge:
    provider: duckdns
    resolvers:
     - "1.1.1.1:53"
     - "8.8.8.8:53"

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo       M-A Set Mark
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo       M-6 Copy
```

```
FileNotFoundError: [Errno 2] No such file or directory: '/etc/dokploy/traefik/treaefik.yml'
ubuntu@nutresa-server:~$ sudo python3 -c 'import yaml; yaml.safe_load(open("/etc/dokploy/traefik/traefik.yml")); print("YAML OK")'
YAML OK
ubuntu@nutresa-server:~$
```

```
ubuntu@nutresa-server:~$ sudo python3 -c 'import yaml; yaml.safe_load(open("/etc/dokploy/traefik/treaefik.yml")); print("YAML OK")'
Traceback (most recent call last):
  File "<string>", line 1, in <module>
FileNotFoundError: [Errno 2] No such file or directory: '/etc/dokploy/traefik/treaefik.yml'
ubuntu@nutresa-server:~$ sudo python3 -c 'import yaml; yaml.safe_load(open("/etc/dokploy/traefik/traefik.yml")); print("YAML OK")'
YAML OK
ubuntu@nutresa-server:~$
```



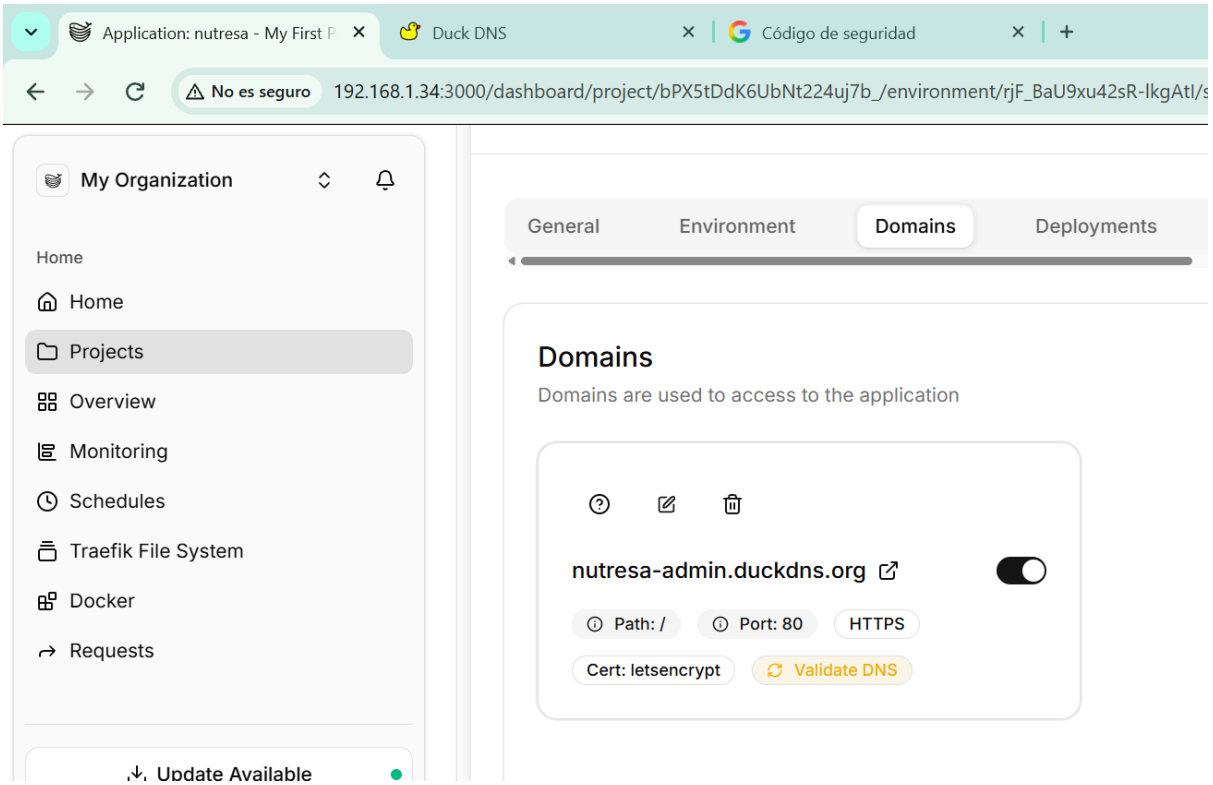

```
ubuntu@nutresa-server:~$ sudo docker ps --format 'table {{.Names}}\t{{.Status}}\t{{.Ports}}' | grep traefik
dokploy-traefik                                          Restarting (1) 14 seconds ago
ubuntu@nutresa-server:~$
```

```
ubuntu@nutresa-server:~$ sudo sed -i 's/insercure/insecure/' /etc/dokploy/traefik/traefik.yml
ubuntu@nutresa-server:~$ sudo grep -n -B1 - A1 'insecure' /etc/dokploy/traefik/traefik.yml
grep: A1: No such file or directory
grep: insecure: No such file or directory
/etc/dokploy/traefik/traefik.yml-9-  watch: true
/etc/dokploy/traefik/traefik.yml:10:  network: dokploy-network
--
/etc/dokploy/traefik/traefik.yml-30-   storage: /etc/dokploy/traefik/dynamic/acme.json
/etc/dokploy/traefik/traefik.yml:31:   caServer: https://acme-staging-v02.api.letsencrypt.org/directory
--
/etc/dokploy/traefik/traefik.yml-34-    resolvers:
/etc/dokploy/traefik/traefik.yml:35:     - "1.1.1.1:53"
/etc/dokploy/traefik/traefik.yml:36:     - "8.8.8.8:53"
ubuntu@nutresa-server:~$ sudo grep -n -B1 -A1 'insecure' /etc/dokploy/traefik/traefik.yml
24-api:
25: insecure: true
26-certificatesResolvers:
ubuntu@nutresa-server:~$ sudo python3 -c 'import yaml; yaml.safe_load(open("/etc/dokploy/traefik/traefik.yml"); print("YAML OK")'
  File "<string>", line 1
    import yaml; yaml.safe_load(open("/etc/dokploy/traefik/traefik.yml"); print("YAML OK")
                                                                        ^
SyntaxError: invalid syntax
ubuntu@nutresa-server:~$ sudo python3 -c 'import yaml; yaml.safe_load(open("/etc/dokploy/traefik/traefik.yml")); print("YAML OK")'
YAML OK
ubuntu@nutresa-server:~$ sudo docker restart dokploy-traefik
dokploy-traefik
ubuntu@nutresa-server:~$ sleep 5
```

```
    resolvers:
     - "1.1.1.1:53"
     - "8.8.8.8:53"
ubuntu@nutresa-server:~$ sudo sed -i 's/exposeByDefault/exposedByDefault/' /etc/dokploy/traefik/traefik.yml
ubuntu@nutresa-server:~$ sudo grep -n 'xposedByDefault\|xposeByDefault' /etc/dokploy/traefik/traefik.yml
5:  exposedByDefault: false
8:  exposedByDefault: false
ubuntu@nutresa-server:~$ sudo python3 -c 'import yaml; yaml.safe_load(open("/etc/dokploy/traefik/traefik.yml")); print("YAML OK")'
YAML OK
ubuntu@nutresa-server:~$ sudo docker restart dokploy-traefik
dokploy-traefik
ubuntu@nutresa-server:~$ sleep 8
ubuntu@nutresa-server:~$ sudo docker logs --since 1m dokploy-traefik
ubuntu@nutresa-server:~$ sudo docker ps --format 'table {{.Names}}\t{{.Status}}\t{{.Ports}}' | grep traefik
dokploy-traefik                                          Up 30 seconds          0.0.0.0:80->80/tcp, [::]:80->80/tcp, 0.0.0.0
:443->443/tcp, [::]:443->443/tcp, 0.0.0.0:443->443/udp, [::]:443->443/udp
ubuntu@nutresa-server:~$
```

```
ubuntu@nutresa-server: ~   Windows PowerShell

ubuntu@nutresa-server:~$ sudo docker logs --since 3m dokploy-traefik
ubuntu@nutresa-server:~$ sudo docker ps --format 'table {{.Names}}\t{{.Status}}' | grep treafik
ubuntu@nutresa-server:~$ curl -sI http://localhost | head -3
HTTP/1.1 404 Not Found
Content-Type: text/plain; charset=utf-8
X-Content-Type-Options: nosniff
ubuntu@nutresa-server:~$
```



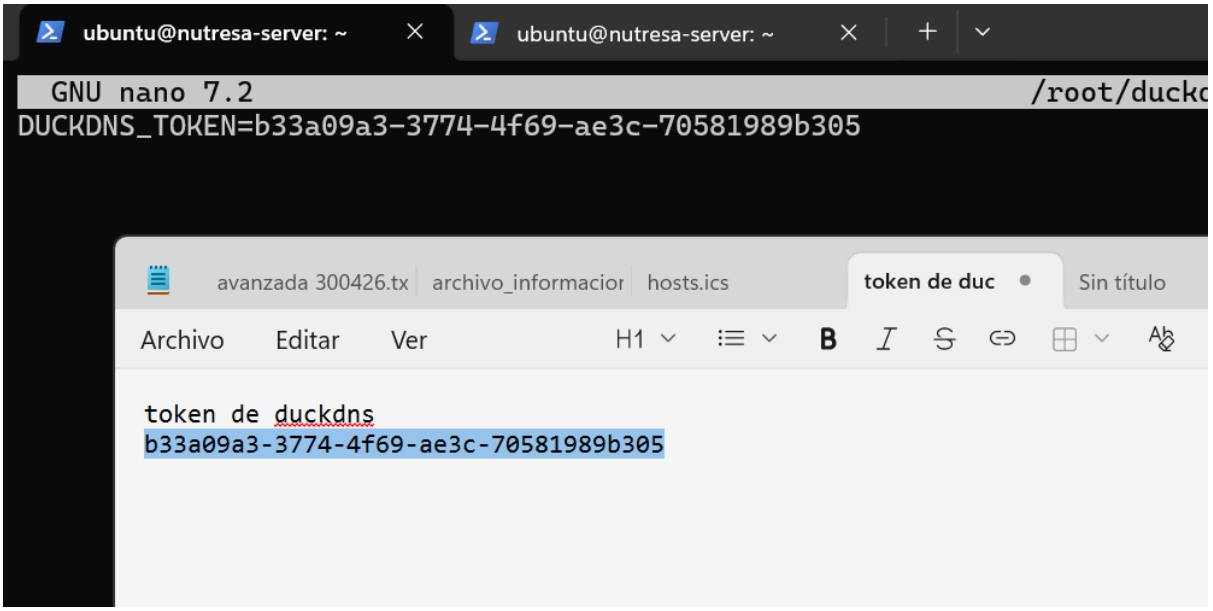

```
ubuntu@nutresa-server:~$ sudo docker inspect dokploy-traefik --format '{{range .Config.Env}}{{println .}}{{end}}' | grep -c DUCKDNS_T
OKEN
1
ubuntu@nutresa-server:~$
```

```
ubuntu@nutresa-server:~$ sudo docker logs -f --since 1m dokploy-traefik 2>&1 | grep -iE 'acme|duckdns|certificate|error|unable'
2026-11-12T21:27:20Z ERR Unable to obtain ACME certificate for domains error="cannot get ACME client duckdns: some credentials inform
ation are missing: DUCKDNS_TOKEN" ACME CA=https://acme-staging-v02.api.letsencrypt.org/directory acmeCA=https://acme-staging-v02.api.
letsencrypt.org/directory domains=["nutresa-admin.duckdns.org"] providerName=letsencrypt.acme routerName=my-first-project-nutresa-qmj
fti-router-websecure-2@file rule=Host(`nutresa-admin.duckdns.org`)
^C
ubuntu@nutresa-server:~$ sudo docker inspect dokploy-traefik --format '{{range .Config.Env}}{{println .}}{{end}}'
DUCKDNS_TOKEN=
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin

ubuntu@nutresa-server:~$ sudo docker inspect dokploy-traefik --format '{{json .NetworkSettings.Networks}}' | grep -o '"dokploy-networ
k"'
"dokploy-network"
ubuntu@nutresa-server:~$
```

```
ubuntu@nutresa-server:~$ read -rsp "b33a09a3-3774-4f69-ae3c-70581989b305" T; echo
b33a09a3-3774-4f69-ae3c-70581989b305
ubuntu@nutresa-server:~$ printf 'DUCKDNS_TOKEN=%s\n' "$T" | sudo tee /root/duckdns.env >/dev/null
ubuntu@nutresa-server:~$ sudo chmod 600 /root/duckdns.env
ubuntu@nutresa-server:~$ unset T
ubuntu@nutresa-server:~$ sudo ls - /root/duckdns.env
ls: cannot access '-': No such file or directory
/root/duckdns.env
ubuntu@nutresa-server:~$ sudo ls -l /root/duckdns.env
-rw------- 1 root root 15 Nov 12 16:29 /root/duckdns.env
ubuntu@nutresa-server:~$ sudo docker rm -f dokploy-traefik
dokploy-traefik
ubuntu@nutresa-server:~$ sudo docker run -d \
> --name dokploy-traefik \
  --restart always \
  --network dokploy-network \
  --env-file /root/duckdns.env \
  -v /etc/dokploy/traefik/traefik.yml:/etc/traefik/traefik.yml \
  -v /etc/dokploy/traefik/dynamic:/etc/dokploy/traefik/dynamic \
  -v /var/run/docker.sock:/var/run/docker.sock:ro \
  -p 80:80/tcp -p 443:443/tcp -p 443:443/udp \
  traefik:v3.6.25
9dc53d42ffeae426756e61f8f71708e1603003666eeeb99918029e715e5c2447
ubuntu@nutresa-server:~$ sudo docker ps --format 'table {{.Names}}\t{{.Status}}\t{{.Ports}}' | grep traefik
dokploy-traefik                                 Up 6 seconds            0.0.0.0:80->80/tcp, [::]:80->80/tcp, 0.0.0.0
:443->443/tcp, [::]:443->443/tcp, 0.0.0.0:443->443/udp, [::]:443->443/udp
ubuntu@nutresa-server:~$ sudo docker inspect dokploy-traefik --format '{{range .Config.Env}}{{println .}}{{end}}' | grep -c DUCKDNS_T
OKEN
1
ubuntu@nutresa-server:~$ sleep 60
```

```
ubuntu@nutresa-server:~$ sudo ls -l /root/duckdns.env
-rw------- 1 root root 15 Nov 12 16:29 /root/duckdns.env
ubuntu@nutresa-server:~$ sudo awk -F= '/^DUCKDNS_TOKEN=/{print "longitud en archivo:", length($2)}' /root/duckdns.env
longitud en archivo: 0
ubuntu@nutresa-server:~$ sudo docker exec dokploy-traefik printenv DUCKDNS_TOKEN | wc -c
1
ubuntu@nutresa-server:~$ sudo docker inspect dokploy-traefik --format 'creado: {{.Created}}'
creado: 2026-11-12T21:30:20.116327628Z
ubuntu@nutresa-server:~$
ubuntu@nutresa-server:~$
```



```
ubuntu@nutresa-server:~$ sudo nano /root/duckdns.env
ubuntu@nutresa-server:~$ sudo chmod 600 /root/duckdns.env
ubuntu@nutresa-server:~$ sudo awk -F= '/^DUCKDNS_TOKEN=/{print "longitud en archivo:", length($2)}' /root/duckdns.env
longitud en archivo: 36
ubuntu@nutresa-server:~$ sudo docker rm -f dokploy-traefik
dokploy-traefik
ubuntu@nutresa-server:~$ sudo docker run -d \
docker: 'docker run' requires at least 1 argument

Usage:  docker run [OPTIONS] IMAGE [COMMAND] [ARG...]

See 'docker run --help' for more information
ubuntu@nutresa-server:~$ sudo docker run -d \
> --name dokploy-traefik \
  --restart always \
  --network dokploy-network \
  --env-file /root/duckdns.env \
  -v /etc/dokploy/traefik/traefik.yml:/etc/traefik/traefik.yml \
  -v /etc/dokploy/traefik/dynamic:/etc/dokploy/traefik/dynamic \
  -v /var/run/docker.sock:/var/run/docker.sock:ro \
  -p 80:80/tcp -p 443:443/tcp -p 443:443/udp \
  traefik:v3.6.25
3d66f9c69f7c3735738d45784d016e3b33d998ab5fbbe2f432a7bca79c26d18f
ubuntu@nutresa-server:~$ sudo docker exec dokploy-traefik printenv DUCKDNS_TOKEN | wc -c
37
ubuntu@nutresa-server:~$ sleep 60
```

```
ubuntu@nutresa-server:~$ sleep 60
ubuntu@nutresa-server:~$ sudo docker logs --since 2m dokploy-traefik 2>&1 | grep -iE 'acme|duckdns|certificate|error|unable'
ubuntu@nutresa-server:~$ echo | openssl s_client -connect 192.168.1.34:443 -servername nutresa-admin.duckdns.org 2>/dev/null | openss
l x509 -noout -issuer -subject -dates

issuer=C = US, O = Let's Encrypt, CN = (STAGING) Dastardly Durum YR1
subject=CN = nutresa-admin.duckdns.org
notBefore=Nov 12 20:37:03 2026 GMT
notAfter=Jan 15 20:37:02 2027 GMT
ubuntu@nutresa-server:~$
ubuntu@nutresa-server:~$
```

ubuntu@nutresa-server: ~ | Windows PowerShell

```
PS C:\Users\Usuario> Test-NetConnection nutresa-admin.duckdns.org -Port 443


ComputerName     : nutresa-admin.duckdns.org
RemoteAddress    : 192.168.1.34
RemotePort       : 443
InterfaceAlias   : vEthernet (Nutresa-External)
SourceAddress    : 192.168.1.14
TcpTestSucceeded : True



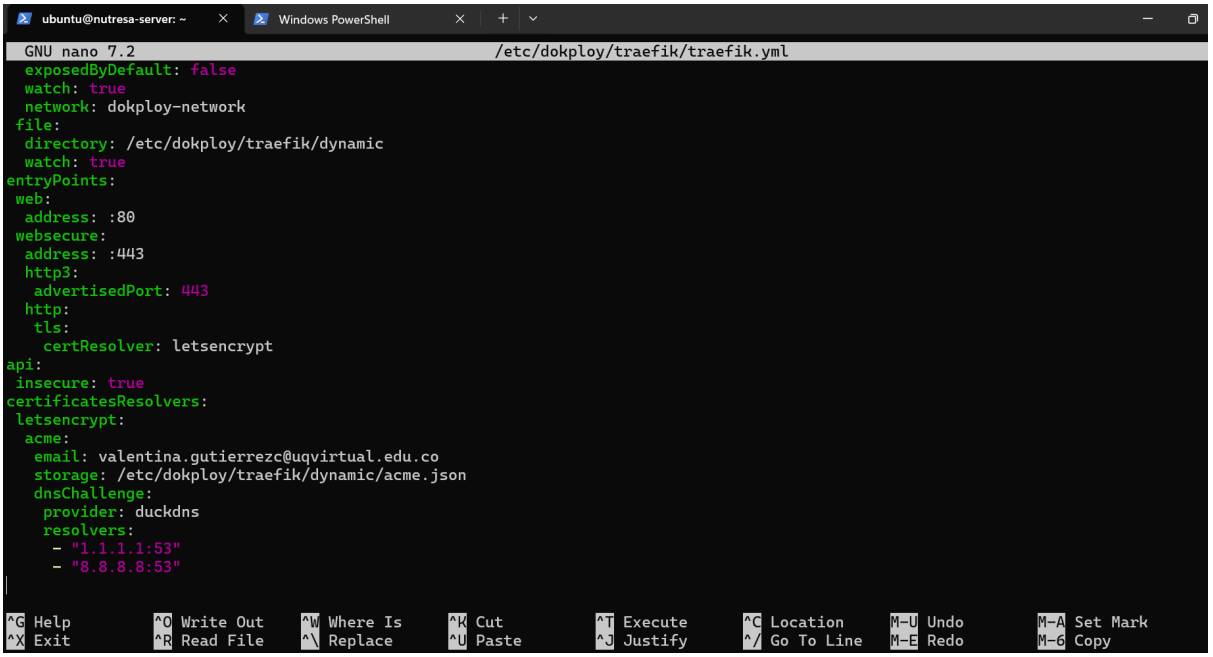
PS C:\Users\Usuario> curl.exe -k -I https://nutresa-admin.duckdns.org
HTTP/1.1 200 OK
Accept-Ranges: bytes
Alt-Svc: h3=":443"; ma=2592000
Content-Length: 2029
Content-Type: text/html
Date: Thu, 12 Nov 2026 21:39:31 GMT
Etag: "6af3faa5-7ed"
Last-Modified: Wed, 11 Nov 2026 05:18:29 GMT
Server: nginx/1.31.8

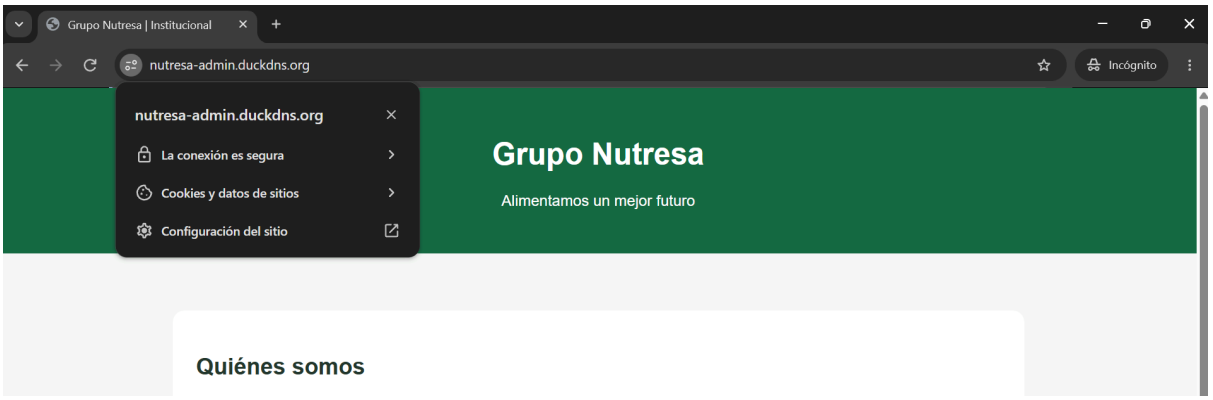
PS C:\Users\Usuario> curl.exe -I http://nutresa-admin.duckdns.org
HTTP/1.1 308 Permanent Redirect
Location: https://nutresa-admin.duckdns.org/
Date: Thu, 12 Nov 2026 21:39:53 GMT
Content-Length: 18

PS C:\Users\Usuario>
```




```
ubuntu@nutresa-server:~$ sudo python3 -c 'import yaml; yaml.safe_load(open("/etc/dokploy/traefik/traefik.yml")); print("YAML OK")'
YAML OK
ubuntu@nutresa-server:~$ sudo docker stop dokploy-traefik
dokploy-traefik
ubuntu@nutresa-server:~$ sudo truncate -s 0 /etc/dokploy/traefik/dynamic/acme.json
ubuntu@nutresa-server:~$ sudo chmod 600 /etc/dokploy/traefik/dynamic/acme.json
ubuntu@nutresa-server:~$ sudo docker start dokploy-traefik
dokploy-traefik
ubuntu@nutresa-server:~$ sleep 60
ubuntu@nutresa-server:~$ sudo docker logs --since 2m dokploy-traefik 2>&1 | grep -iE 'acme|duckdns|certificate|error|unable'
ubuntu@nutresa-server:~$ echo | openssl s_client -connect 192.168.1.34:443 -servername nutresa-admin.duckdns.org 2>/dev/null | openss
l x509 -noout -issuer -subject -dates
issuer=C = US, O = Let's Encrypt, CN = YR2
subject=CN = nutresa-admin.duckdns.org
notBefore=Nov 12 20:44:36 2026 GMT
notAfter=Feb 10 20:44:35 2027 GMT
ubuntu@nutresa-server:~$
```


ubuntu@nutresa-server: ~
Windows PowerShell
GNU nano 7.2 /etc/dokploy/traefik/traefik.yml
exposedByDefault: false
watch: true
network: dokploy-network
file:
directory: /etc/dokploy/traefik/dynamic
watch: true
entryPoints:
web:
address: :80
websecure:
address: :443
http3:
advertisedPort: 443
http:
tls:
certResolver: letsencrypt
api:
insecure: true
certificatesResolvers:
letsencrypt:
acme:
email: valentina.gutierrezc@uqvirtual.edu.co
storage: /etc/dokploy/traefik/dynamic/acme.json
dnsChallenge:
provider: duckdns
resolvers:
- "1.1.1.1:53"
- "8.8.8.8:53"
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy


Grupo Nutresa | Institucional
nutresa-admin.duckdns.org
Incógnito
nutresa-admin.duckdns.org
La conexión es segura
Cookies y datos de sitios
Configuración del sitio
Grupo Nutresa
Alimentamos un mejor futuro
Quiénes somos

SEMANA 3

```
C:\Users\user\OneDrive - uqvirtual.edu.co>git clone https://github.com/ValRodriguez97/Dise-o_Infra
estructura_TI_Nutresa.git
Cloning into 'Dise-o_Infraestructura_TI_Nutresa'...
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 7 (delta 0), reused 4 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (7/7), 18.46 KiB | 378.00 KiB/s, done.

C:\Users\user\OneDrive - uqvirtual.edu.co>cd Dise-o_Infraestructura_TI_Nutresa

C:\Users\user\OneDrive - uqvirtual.edu.co\Dise-o_Infraestructura_TI_Nutresa
C:\Users\user\OneDrive - uqvirtual.edu.co\Dise-o_Infraestructura_TI_Nutresa
C:\Users\user\OneDrive - uqvirtual.edu.co\Dise-o_Infraestructura_TI_Nutresa>git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

C:\Users\user\OneDrive - uqvirtual.edu.co\Dise-o_Infraestructura_TI_Nutresa>dir
 El volumen de la unidad C no tiene etiqueta.
 El número de serie del volumen es: E40F-A067

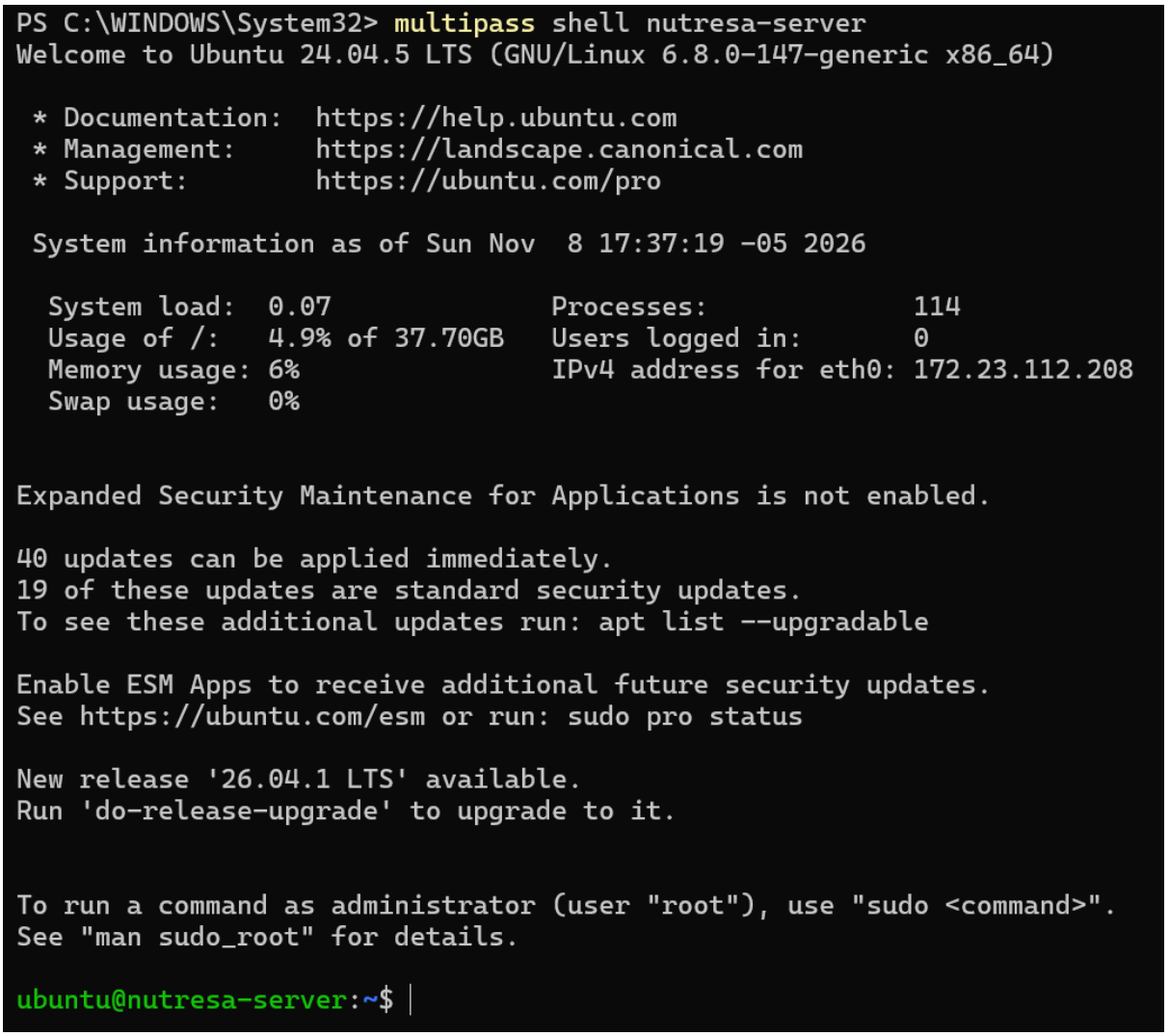
 Directorio de C:\Users\user\OneDrive - uqvirtual.edu.co\Dise-o_Infraestructura_TI_Nutresa

08/11/2026  04:19 p. m.    <DIR>          .
08/11/2026  04:19 p. m.    <DIR>          ..
08/11/2026  04:19 p. m.            15.157 analisis.md
08/11/2026  04:19 p. m.               278 diagrama.mmd
08/11/2026  04:19 p. m.            35.823 LICENSE
               3 archivos         51.258 bytes
               2 dirs  110.484.086.784 bytes libres
```

Se crea la máquina virtual

```
PS C:\WINDOWS\System32> multipass launch 24.04 --name nutresa-server --cpus 2 --memory 4G --disk 40G
Launched: nutresa-server
PS C:\WINDOWS\System32> multipass list
Name                    State             IPv4             Image
nutresa-server          Running           172.23.112.208   Ubuntu 24.04 LTS
prueba-ip               Stopped           --               Ubuntu 22.04 LTS
roma                    Unknown           --               Ubuntu 22.04 LTS
PS C:\WINDOWS\System32> multipass info nutresa-server
Name:           nutresa-server
State:          Running
Snapshots:      0
IPv4:           172.23.112.208
Release:        Ubuntu 24.04.5 LTS
Image hash:     04e8f2461a16 (Ubuntu 24.04 LTS)
CPU(s):         2
Load:           0.11 0.16 0.07
Disk usage:     1.9GiB out of 38.7GiB
Memory usage:   438.3MiB out of 3.8GiB
Mounts:         --
```

Se ingresa a la maquina virtual



Se actualiza la maquina y sus paquetes

```
ubuntu@nutresa-server:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
34 packages can be upgraded. Run 'apt list --upgradable' to see
them.
ubuntu@nutresa-server:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following NEW packages will be installed:
  linux-headers-6.8.0-149 linux-headers-6.8.0-149-generic
  linux-image-6.8.0-149-generic
  linux-modules-6.8.0-149-generic linux-tools-6.8.0-149
  linux-tools-6.8.0-149-generic
The following packages will be upgraded:
  apt apt-utils bind9-dnsutils bind9-host bind9-libs
  ca-certificates cloud-init distro-info-data gir1.2-glib-2.0
  git git-man libapt-pkg6.0t64 libbrotli1 libglib2.0-0t64
```

se descargan las herramientas Estas herramientas nos servirán para administrar el servidor, probar la conectividad, configurar servicios y gestionar la seguridad.

```
ubuntu@nutresa-server:~$ sudo apt install -y curl wget git nano
vim net-tools iproute2 dnsutils iputils-ping traceroute chrony o
penssh-server ufw
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.16).
curl set to manually installed.
wget is already the newest version (1.21.4-1ubuntu4.5).
wget set to manually installed.
git is already the newest version (1:2.43.0-1ubuntu7.4).
git set to manually installed.
nano is already the newest version (7.2-2ubuntu0.2).
nano set to manually installed.
vim is already the newest version (2:9.1.0016-1ubuntu7.20).
vim set to manually installed.
iproute2 is already the newest version (6.1.0-1ubuntu6.4).
iproute2 set to manually installed.
iputils-ping is already the newest version (3:20240117-1ubuntu0.
1).
iputils-ping set to manually installed.
openssh-server is already the newest version (1:9.6p1-3ubuntu13.
20).
ufw is already the newest version (0.36.2-6).
ufw set to manually installed.
The following additional packages will be installed:
  tzdata-legacy
The following packages will be REMOVED:
  systemd-timesyncd
The following NEW packages will be installed:
  chrony dnsutils net-tools traceroute tzdata-legacy
```

Se configuran nobre y se verifican ip puerta y servidor?

```
ubuntu@nutresa-server:~$ sudo hostnamectl set-hostname nutresa-s
erver
ubuntu@nutresa-server:~$ hostnamectl
 Static hostname: nutresa-server
       Icon name: computer-vm
         Chassis: vm
      Machine ID: 975c1ca04cb946deb51effde025597b1
         Boot ID: 15c2d45a3e6648d093acab9327089def
  Virtualization: microsoft
Operating System: Ubuntu 24.04.5 LTS
          Kernel: Linux 6.8.0-147-generic
    Architecture: x86-64
 Hardware Vendor: Microsoft Corporation
  Hardware Model: Virtual Machine
Firmware Version: Hyper-V UEFI Release v4.1
   Firmware Date: Thu 2025-09-25
    Firmware Age: 1y 1month 2w
ubuntu@nutresa-server:~$ ip -br addr
lo               UNKNOWN        127.0.0.1/8 ::1/128
eth0             UP             172.23.112.208/20 metric 100 fe8
0::5054:ff:fe26:3e0f/64
ubuntu@nutresa-server:~$ ip route
default via 172.23.112.1 dev eth0 proto dhcp src 172.23.112.208
metric 100
172.23.112.0/20 dev eth0 proto kernel scope link src 172.23.112.
208 metric 100
172.23.112.1 dev eth0 proto dhcp scope link src 172.23.112.208 m
etric 100
```

Se comprueba internet ? con ping

```
ubuntu@nutresa-server:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=20.6 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=118 time=62.7 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=118 time=20.9 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=118 time=107 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 20.585/52.825/107.166/35.739 ms
ubuntu@nutresa-server:~$ ping -c 4 google.com
PING google.com (172.217.30.174) 56(84) bytes of data.
64 bytes from eze03s36-in-f14.1e100.net (172.217.30.174): icmp_s
eq=1 ttl=117 time=22.7 ms
64 bytes from eze03s36-in-f14.1e100.net (172.217.30.174): icmp_s
eq=2 ttl=117 time=18.6 ms
64 bytes from eze03s36-in-f14.1e100.net (172.217.30.174): icmp_s
eq=3 ttl=117 time=66.4 ms
64 bytes from eze03s36-in-f14.1e100.net (172.217.30.174): icmp_s
eq=4 ttl=117 time=88.5 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 18.623/49.087/88.542/29.495 ms
```

SSH

```
ubuntu@nutresa-server:~$ sudo systemctl enable --now ssh
Synchronizing state of ssh.service with SysV service script with
 /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/sshd.service → /usr/lib/syst
emd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.
service → /usr/lib/systemd/system/ssh.service.
ubuntu@nutresa-server:~$ sudo systemctl status ssh --no-pager
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enable
d; preset: enabled)
     Active: active (running) since Sun 2026-11-08 17:42:17 -05;
 7min ago
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 12021 (sshd)
      Tasks: 1 (limit: 4634)
     Memory: 1.2M (peak: 1.6M)
        CPU: 42ms
     CGroup: /system.slice/ssh.service
             └─12021 "sshd: /usr/sbin/sshd -D [listener] 0 of 1…

Nov 08 17:42:17 nutresa-server sshd[12021]: Server listening …2.
Nov 08 17:42:17 nutresa-server systemd[1]: Started ssh.servic…r.
Nov 08 17:42:17 nutresa-server sshd[12021]: Server listening …2.
Hint: Some lines were ellipsized, use -l to show in full.
ubuntu@nutresa-server:~$
```

Error en ip ya que esta en 172.20.176.1 y no esta dentro de la red de wifi y va a dificultar las pruebas

```
PS C:\WINDOWS\System32> multipass networks
Name                    Type     Description
Default Switch          switch   Virtual Switch with internal n
etworking
WSL (Hyper-V firewall)  switch   Virtual Switch with internal n
etworking
PS C:\WINDOWS\System32> Get-NetAdapter | Format-Table Name, Inte
rfaceDescription, Status

Name                                InterfaceDescription
----                                --------------------
Conexión de red Bluetooth           Bluetooth Device (Persona...
vEthernet (WSL (Hyper-V firewall)) Hyper-V Virtual Ethernet ...
Ethernet 3                          VirtualBox Host-Only Ethe...
Wi-Fi                               Realtek RTL8852AE WiFi 6 ...
vEthernet (Default Switch)          Hyper-V Virtual Ethernet ...


PS C:\WINDOWS\System32> Get-VMSwitch | Format-Table Name, Switch
Type, NetAdapterInterfaceDescription
Get-VMSwitch : No dispone del permiso necesario para completar
esta tarea. Póngase en contacto con el administrador de la
directiva de autorización del equipo 'VALENTINA-VMOI4'.
En línea: 1 Carácter: 1
+ Get-VMSwitch | Format-Table Name, SwitchType,
NetAdapterInterfaceDesc ...
+ ~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: (:) [Get-VMSwitch]
   , VirtualizationException
    + FullyQualifiedErrorId : Unspecified,Microsoft.HyperV.Pow
   erShell.Commands.GetVMSwitch
```

```
PS C:\WINDOWS\system32> Get-NetIPConfiguration


InterfaceAlias        : vEthernet (Default Switch)
InterfaceIndex        : 24
InterfaceDescription  : Hyper-V Virtual Ethernet Adapter
IPv4Address           : 172.23.112.1
IPv6DefaultGateway    :
IPv4DefaultGateway    :
DNSServer             : fec0:0:0:ffff::1
                        fec0:0:0:ffff::2
                        fec0:0:0:ffff::3

InterfaceAlias        : vEthernet (WSL (Hyper-V firewall))
InterfaceIndex        : 65
InterfaceDescription  : Hyper-V Virtual Ethernet Adapter #2
IPv4Address           : 172.20.176.1
IPv6DefaultGateway    :
IPv4DefaultGateway    :
```

Se ejecutaron 4 comandos
Get-VMSwitch | Format-List *
Get-NetIPConfiguration
Get-NetAdapter -Name "Wi-Fi" | Format-List *
multipass networks

```
IPv4DefaultGateway    :
DNSServer             : fec0:0:0:ffff::1
                        fec0:0:0:ffff::2
                        fec0:0:0:ffff::3

InterfaceAlias        : Ethernet 3
InterfaceIndex        : 9
InterfaceDescription  : VirtualBox Host-Only Ethernet Adapter
IPv4Address           : 192.168.56.1
IPv6DefaultGateway    :
IPv4DefaultGateway    :
DNSServer             : fec0:0:0:ffff::1
                        fec0:0:0:ffff::2
                        fec0:0:0:ffff::3

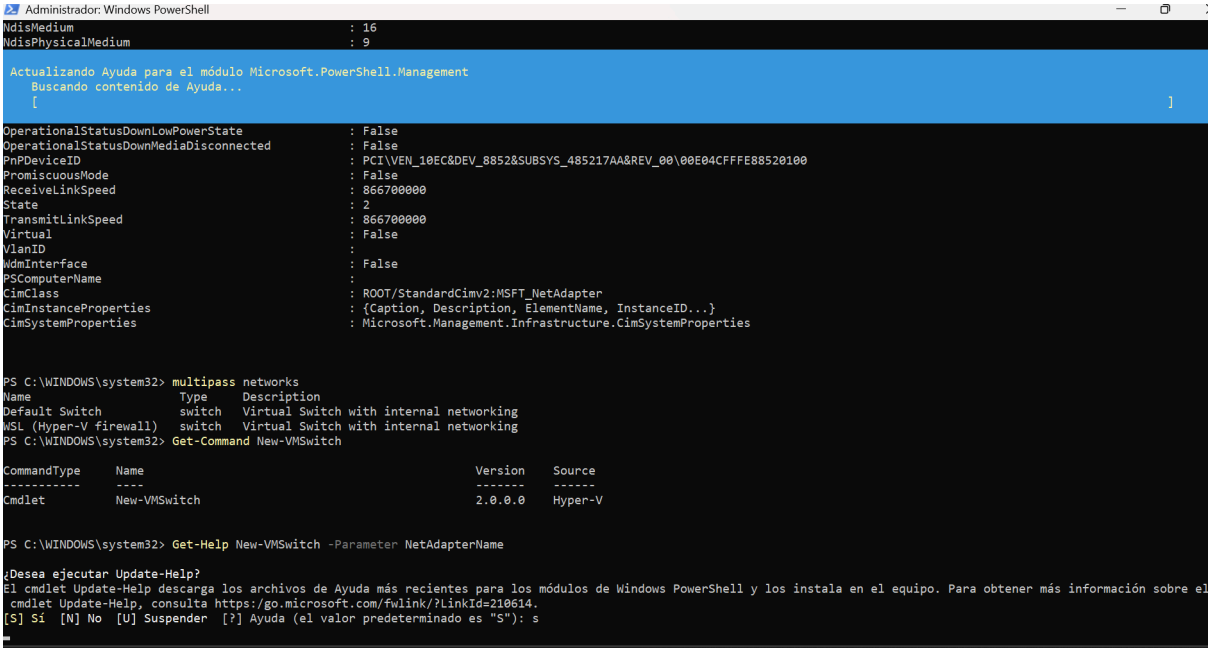
InterfaceAlias        : Wi-Fi
InterfaceIndex        : 6
InterfaceDescription  : Realtek RTL8852AE WiFi 6 802.11ax PCIe Adapter
NetProfile.Name       : FLIA SOLORZANO 2.4
IPv4Address           : 192.168.1.31
IPv6DefaultGateway    :
IPv4DefaultGateway    : 192.168.1.1
DNSServer             : 200.21.200.10
                        200.21.200.80

InterfaceAlias        : Conexión de red Bluetooth
InterfaceIndex        : 21
InterfaceDescription  : Bluetooth Device (Personal Area Network)
NetAdapter.Status     : Disconnected

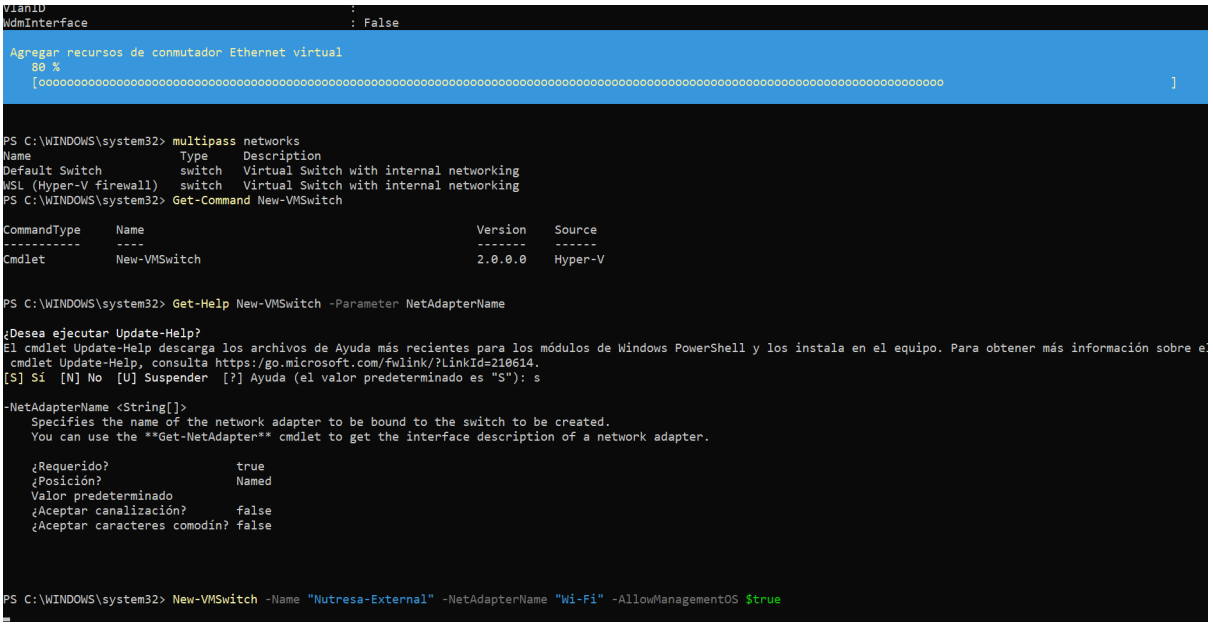

PS C:\WINDOWS\system32> Get-VMSwitch | Format-List *


DefaultQueueVmmqQueuePairs                     : 0
DefaultQueueVmmqQueuePairsRequested            : 16
Name                                           : Default Switch
Id                                             : c08cb7b8-9b3c-408e-8e30-5e16a3aeb444
Notes                                          :
Extensions                                     : {Microsoft Azure VFP Switch Filter Extension, Plataforma de
                                                 filtrado de Microsoft Windows, Captura de NDIS de Microsoft}
```

Consultamos la posibilidad de crear un switch externo con los dos ultimos comando



Se crea un switch




```
PS C:\WINDOWS\system32> New-VMSwitch -Name "Nutresa-External" -NetAdapterName "Wi-Fi" -AllowManagementOS $true

Name              SwitchType NetAdapterInterfaceDescription
----              ---------- ------------------------------
Nutresa-External External   Realtek RTL8852AE WiFi 6 802.11ax PCIe Adapter


PS C:\WINDOWS\system32>
```

se comprueba exitosamente

```
PS C:\WINDOWS\system32> Get-VMSwitch

Name                  SwitchType NetAdapterInterfaceDescription
----                  ---------- ------------------------------
Default Switch        Internal
WSL (Hyper-V firewall) Internal
Nutresa-External      External   Realtek RTL8852AE WiFi 6 802.11ax PCIe Adapter


PS C:\WINDOWS\system32> multipass networks
Name                    Type     Description
Default Switch          switch   Virtual Switch with internal networking
Nutresa-External        switch   Virtual Switch with external networking
WSL (Hyper-V firewall)  switch   Virtual Switch with internal networking
PS C:\WINDOWS\system32>
```

para asignar el switch se detuvo la maquina

```
PS C:\WINDOWS\System32> multipass stop nutresa-server
PS C:\WINDOWS\System32> multipass set local.bridged-network=Nutr
esa-External
PS C:\WINDOWS\System32> multipass set local.nutresa-server.bridg
ed=true
PS C:\WINDOWS\System32> multipass start nutresa-server

############################################################################
#########################
Multipass 1.17.0 release
Fully open source, custom image launch, GUI improvements, and ma
ny fixes.

Go here for more information: https://github.com/canonical/multi
pass/releases/tag/v1.17.0
############################################################################
#########################
PS C:\WINDOWS\System32> multipass list
Name                    State             IPv4             Image
nutresa-server          Running           172.23.115.116   Ubunt
u 24.04 LTS
                                          192.168.1.31
prueba-ip               Stopped           --               Ubunt
u 22.04 LTS
roma                    Unknown           --               Ubunt
u 22.04 LTS
PS C:\WINDOWS\System32> multipass shell nutresa-server
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 6.8.0-149-generic x86_6
4)
```

y se confirmo que si se cambio, surgio otro problema, tiene la msima direccion ip que el computador donde se cre la maquina virtual

```
ubuntu@nutresa-server:~$ ip -br address
lo               UNKNOWN        127.0.0.1/8 ::1/128
eth0             UP             172.23.115.116/20 metric 100 fe8
0::5054:ff:fe26:3e0f/64
eth1             UP             192.168.1.31/24 metric 200 fe80:
:5054:ff:fedf:74c0/64
ubuntu@nutresa-server:~$ exit
logout
PS C:\WINDOWS\System32> multipass stop nutresa-server
PS C:\WINDOWS\System32>
```

ahora si se le asigno una direccion ip valida a la MV

```
ubuntu@nutresa-server:~$ sudo arping -D -I eth1 -c 4 192.168.1.1
0
ARPING 192.168.1.10 from 0.0.0.0 eth1
Sent 4 probes (4 broadcast(s))
Received 0 response(s)
ubuntu@nutresa-server:~$ sudo ip -4 addr flush dev eth1
ubuntu@nutresa-server:~$ sudo ip addr add 192.168.1.10/24 dev et
h1
ubuntu@nutresa-server:~$ sudo ip link set eth1 up
ubuntu@nutresa-server:~$ ip -4 -br address
lo               UNKNOWN        127.0.0.1/8
eth0             UP             172.23.127.202/20 metric 100
eth1             UP             192.168.1.10/24
ubuntu@nutresa-server:~$
```

Segundo intento
se verifica la red/switch creado

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> Get-VMSwitch | Format-Table Name, SwitchType, NetAdapterInterfaceDescription

Name            SwitchType NetAdapterInterfaceDescription
----            ---------- ------------------------------
Nutresa-External   External Realtek RTL8852AE WiFi 6 802.11ax PCIe Adapter
Default Switch     Internal


PS C:\WINDOWS\system32> Get-NetAdapter | Format-Table Name, InterfaceDescription, Status

Name                      InterfaceDescription                      Status
----                      --------------------                      ------
Conexión de red Bluetooth    Bluetooth Device (Personal Area Network)       Disconnected
vEthernet (Default Switch)   Hyper-V Virtual Ethernet Adapter               Up
vEthernet (Nutresa-External) Hyper-V Virtual Ethernet Adapter #3           Up
Ethernet 3                   VirtualBox Host-Only Ethernet Adapter         Up
Wi-Fi                        Realtek RTL8852AE WiFi 6 802.11ax PCIe Adapter Up
Puente de red                Microsoft Network Adapter Multiplexor Driver   Up
```

Creación del cliente servidor en ubuntu con multipass

```
PS C:\WINDOWS\system32> multipass list
No instances found.
PS C:\WINDOWS\system32> multipass get local.driver
virtualbox
PS C:\WINDOWS\system32> multipass launch 24.04 --name nutresa-cliente --cpus 2 --memory 2G --disk 20G
Launched: nutresa-cliente
PS C:\WINDOWS\system32> multipass shell nutresa-cliente
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 6.8.0-147-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Nov 10 16:08:09 -05 2026

  System load:             0.4
  Usage of /:              10.1% of 18.33GB
  Memory usage:            10%
  Swap usage:              0%
  Processes:               124
  Users logged in:         0
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:5054:ff:fef3:24a3


Expanded Security Maintenance for Applications is not enabled.

45 updates can be applied immediately.
24 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '26.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@nutresa-cliente:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

Actualizando y descarga de herramientas (se prepara el cliente)

```
ubuntu@nutresa-cliente:~$ sudo apt install -y openssh-client nfs-common nis rpcbind samba-client cups-client chrony
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'smbclient' instead of 'samba-client'
openssh-client is already the newest version (1:9.6p1-3ubuntu13.20).
openssh-client set to manually installed.
The following additional packages will be installed:
  cups-common keyutils libavahi-client3 libavahi-common-data libavahi-common3 libcups2t64 libldb2 libnfsidmap1 libnsl2
  libnss-nis libsmbclient0 libtalloc2 libtdb1 libtevent0t64 libwbclient0 make nscd python3-gpg python3-ldb
  python3-markdown python3-samba python3-talloc python3-tdb samba-common samba-common-bin samba-dsdb-modules
  samba-libs tzdata-legacy yp-tools ypbind-mt ypserv
Suggested packages:
  cups cups-bsd make-doc watchdog python-markdown-doc heimdal-clients python3-dnspython cifs-utils krb5-kdc
The following packages will be REMOVED:
  systemd-timesyncd
The following NEW packages will be installed:
  chrony cups-client cups-common keyutils libavahi-client3 libavahi-common-data libavahi-common3 libcups2t64 libldb2
  libnfsidmap1 libnsl2 libnss-nis libsmbclient0 libtalloc2 libtdb1 libtevent0t64 libwbclient0 make nfs-common nis nscd
  python3-gpg python3-ldb python3-markdown python3-samba python3-talloc python3-tdb rpcbind samba-common
  samba-common-bin samba-dsdb-modules samba-libs smbclient tzdata-legacy yp-tools ypbind-mt ypserv
0 upgraded, 37 newly installed, 1 to remove and 3 not upgraded.
Need to get 14.0 MB of archives.
After this operation, 65.4 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnfsidmap1 amd64 1:2.6.4-3ubuntu5.1 [48.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 rpcbind amd64 1.2.6-7ubuntu2 [46.5 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 keyutils amd64 1.6.3-3build1 [56.8 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nfs-common amd64 1:2.6.4-3ubuntu5.1 [248 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 nscd amd64 2.39-0ubuntu8.9 [70.5 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 tzdata-legacy all 2026e-0ubuntu0.24.04.1 [95.4 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 chrony amd64 4.5-1ubuntu4.2 [316 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 cups-common all 2.4.7-1.2ubuntu7.14 [264 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libavahi-common-data amd64 0.8-13ubuntu6.2 [30.1 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libavahi-common3 amd64 0.8-13ubuntu6.2 [23.4 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libavahi-client3 amd64 0.8-13ubuntu6.2 [26.8 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcups2t64 amd64 2.4.7-1.2ubuntu7.14 [274 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 cups-client amd64 2.4.7-1.2ubuntu7.14 [154 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble/main amd64 libtalloc2 amd64 2.4.2-1build2 [27.3 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble/main amd64 libtdb1 amd64 1.4.10-1build1 [46.8 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble/main amd64 libtevent0t64 amd64 0.16.1-2build1 [42.6 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libldb2 amd64 2:2.8.0+samba4.19.5+dfsg-4ubuntu9.8 [188 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble/main amd64 libnsl2 amd64 1.3.0-3build3 [41.4 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble/universe amd64 libnss-nis amd64 3.1-0ubuntu7 [26.5 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libwbclient0 amd64 2:4.19.5+dfsg-4ubuntu9.8 [72.3 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 samba-libs amd64 2:4.19.5+dfsg-4ubuntu9.8 [6019 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libsmbclient0 amd64 2:4.19.5+dfsg-4ubuntu9.8 [62.4 kB]
```

comunicacion entre maquinas

```
ubuntu@nutresa-cliente:~$ ping -c 4 192.168.1.34
PING 192.168.1.34 (192.168.1.34) 56(84) bytes of data.
64 bytes from 192.168.1.34: icmp_seq=1 ttl=255 time=70.7 ms
64 bytes from 192.168.1.34: icmp_seq=2 ttl=255 time=64.2 ms
64 bytes from 192.168.1.34: icmp_seq=3 ttl=255 time=11.0 ms
64 bytes from 192.168.1.34: icmp_seq=4 ttl=255 time=11.6 ms

--- 192.168.1.34 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 11.015/39.382/70.706/28.153 ms
```

Instalación desde nutresa-server de NIS (primero se actualizo y luego se descargo NIS y sus herramientas).

```
ubuntu@nutresa-server:~$ ip link show eth1
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 52:54:00:1d:99:07 brd ff:ff:ff:ff:ff:ff
ubuntu@nutresa-server:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@nutresa-server:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following upgrades have been deferred due to phasing:
  distro-info python3-distro-info snapd
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
ubuntu@nutresa-server:~$ sudo apt install -y chrony nfs-kernel-server rpcbind nis samba cups
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  acl attr avahi-daemon colord colord-data cups-browsed cups-client cups-common cups-core-drivers cups-daemon cups-filters
  cups-filters-core-drivers cups-ipp-utils cups-ppdc cups-server-common dconf-gsettings-backend dconf-service fonts-droid-fallback
  fonts-noto-mono fonts-urw-base35 ghostscript glib-networking glib-networking-common glib-networking-services
  gsettings-desktop-schemas ipp-usb keyutils libavahi-client3 libavahi-common-data libavahi-common3 libavahi-core7 libavahi-glib1
  libboost-iostreams1.83.0 libboost-locale1.83.0 libboost-thread1.83.0 libcairo2 libcephfs2 libcolord2 libcolorhug2 libcups2t64
  libcupsfilters2-common libcupsfilters2t64 libdaemon0 libdconf1 libexif12 libfontenc1 libgphoto2-6t64 libgphoto2-l10n
  libgphoto2-port12t64 libgs-common libgs10 libgs10-common libgusb2 libice6 libidn12 libieee1284-3t64 libijs-0.35 libjbig2dec0
  liblcms2-2 libldb2 liblouis-data liblouis20 liblouisutdml-bin liblouisutdml-data liblouisutdml9t64 libltdl7 libnfsidmap1 libnsl2
```

IMPLEMENTACIÓN DE CHRONY / NTP Y NFS

instalado los paquetes básicos

```
ubuntu@nutresa-server:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1445 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1711 kB]
Fetched 3282 kB in 3s (1255 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@nutresa-server:~$ sudo apt install -y chrony
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
chrony is already the newest version (4.5-1ubuntu4.2).
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
ubuntu@nutresa-server:~$
```

Se activa en servicio y verifica en estado de CHRONY

```
ubuntu@nutresa-server:~$ sudo systemctl enable --now chrony
Synchronizing state of chrony.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable chrony
ubuntu@nutresa-server:~$ sudo systemctl status chrony
● chrony.service - chrony, an NTP client/server
     Loaded: loaded (/usr/lib/systemd/system/chrony.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-11-10 16:32:43 -05; 1h 16min ago
       Docs: man:chronyd(8)
             man:chronyc(1)
             man:chrony.conf(5)
   Main PID: 4614 (chronyd)
      Tasks: 2 (limit: 4614)
     Memory: 1.4M (peak: 2.1M)
        CPU: 150ms
     CGroup: /system.slice/chrony.service
             ├─4614 /usr/sbin/chronyd -F 1
             └─4615 /usr/sbin/chronyd -F 1

Nov 10 16:32:43 nutresa-server systemd[1]: Starting chrony.service - chrony, an NTP client/server...
Nov 10 16:32:43 nutresa-server chronyd[4614]: chronyd version 4.5 starting (+CMDMON +NTP +REFCLOCK +RTC +PRIVDROP +SCFILTER +SIGND +>
Nov 10 16:32:43 nutresa-server chronyd[4614]: Loaded 0 symmetric keys
Nov 10 16:32:43 nutresa-server chronyd[4614]: Initial frequency 5.944 ppm
Nov 10 16:32:43 nutresa-server chronyd[4614]: Using right/UTC timezone to obtain leap second data
Nov 10 16:32:43 nutresa-server chronyd[4614]: Loaded seccomp filter (level 1)
Nov 10 16:32:43 nutresa-server systemd[1]: Started chrony.service - chrony, an NTP client/server.
Nov 10 16:32:49 nutresa-server chronyd[4614]: Selected source 91.189.91.157 (ntp.ubuntu.com)
Nov 10 16:32:49 nutresa-server chronyd[4614]: System clock TAI offset set to 37 seconds
Nov 10 16:33:54 nutresa-server chronyd[4614]: Selected source 38.19.83.110 (0.ubuntu.pool.ntp.org)

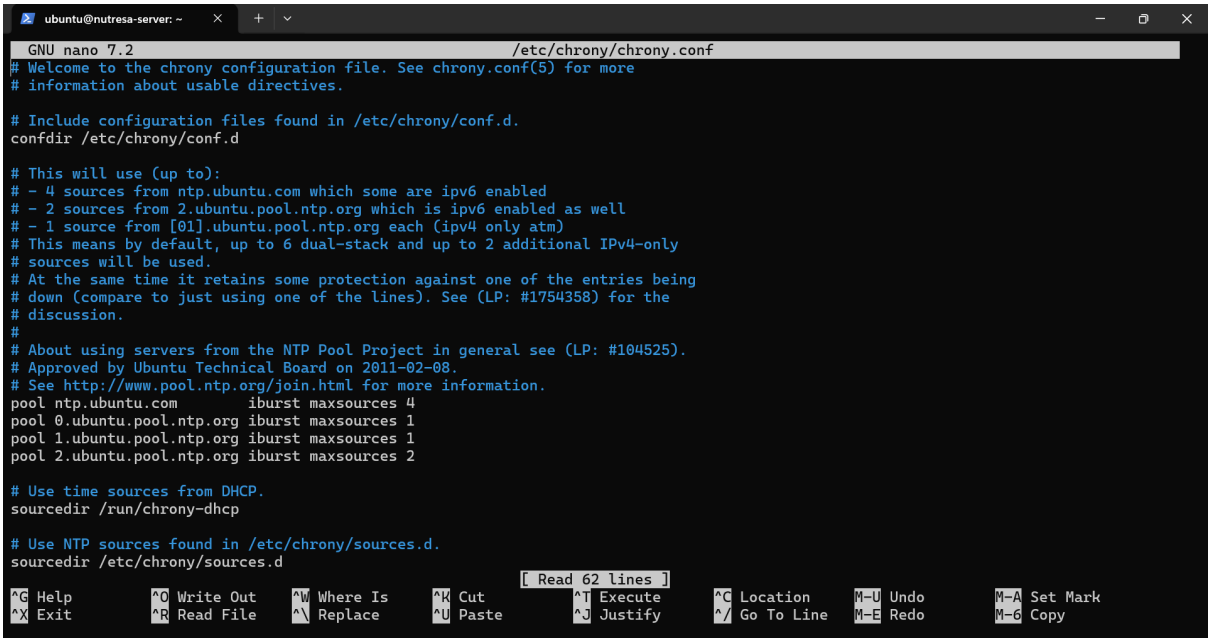
ubuntu@nutresa-server:~$
```

al final de nano se agrego “

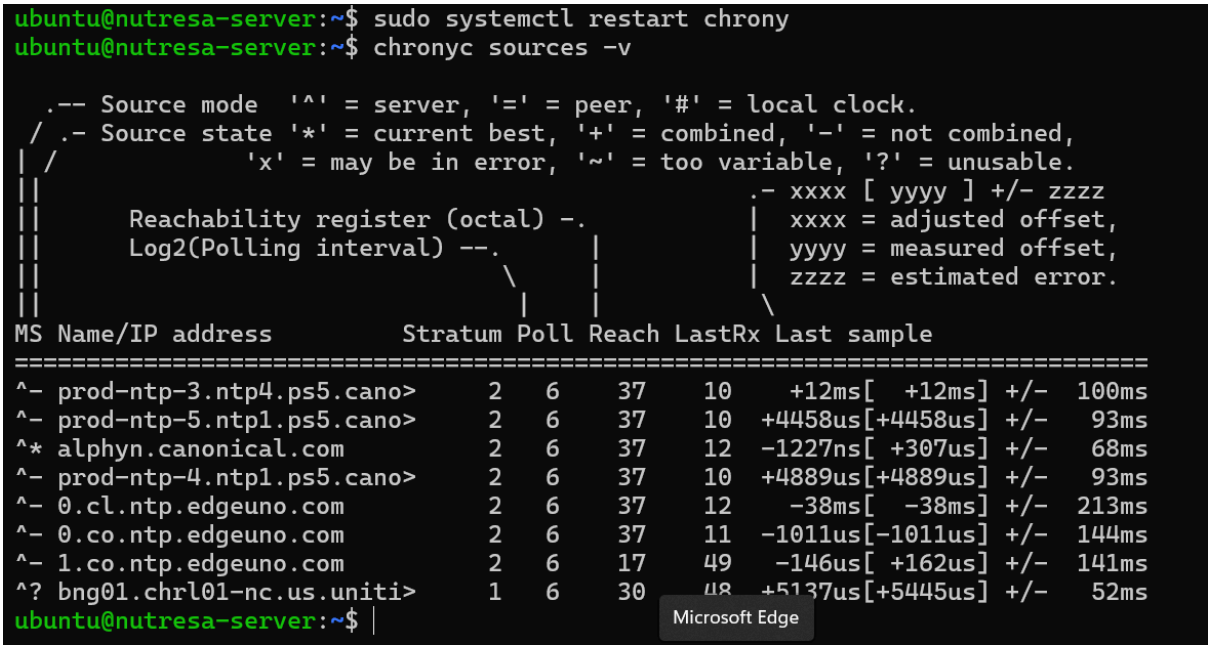
```
None
# Servidor NTP de Nutresa
allow 192.168.1.0/24

# Permitir sincronizacion de clientes de la red local
```

“ La directiva allow permite que los equipos de la red local consulten la hora del servidor.



Se reinicia y se comprueba sus fuentes



chronyc tracking - se comprueba el estado de sincronizacion

```
ubuntu@nutresa-server:~$ chronyc tracking
Reference ID    : 5BBD5B9D (alphyn.canonical.com)
Stratum         : 3
Ref time (UTC)  : Tue Nov 10 22:55:12 2026
System time     : 0.000137431 seconds fast of NTP time
Last offset     : +0.000140441 seconds
RMS offset      : 0.001591483 seconds
Frequency       : 30.339 ppm fast
Residual freq   : +4.602 ppm
Skew            : 1.496 ppm
Root delay      : 0.132102475 seconds
Root dispersion : 0.002445091 seconds
Update interval : 64.2 seconds
Leap status     : Normal
ubuntu@nutresa-server:~$
```

Si está activo, permite el puerto NTP

```
ubuntu@nutresa-server:~$ sudo ufw allow from 192.168.1.0/24 to any port 123 proto udp
Rules updated
ubuntu@nutresa-server:~$
```

Cliente

```
ubuntu@nutresa-cliente:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1445 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1711 kB]
Fetched 3282 kB in 3s (1011 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@nutresa-cliente:~$ sudo apt install -y chrony
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
chrony is already the newest version (4.5-1ubuntu4.2).
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
ubuntu@nutresa-cliente:~$ sudo nano /etc/chrony/chrony.conf
```

```
ubuntu@nutresa-cliente:~$ sudo systemctl enable --now chrony
Synchronizing state of chrony.service with SysV service script with /usr/lib/systemd/systemd-sysv-instal
l.
Executing: /usr/lib/systemd/systemd-sysv-install enable chrony
ubuntu@nutresa-cliente:~$ sudo systemctl restart chrony
ubuntu@nutresa-cliente:~$ chronyc sources -v

  .-- Source mode  '^' = server, '=' = peer, '#' = local clock.
 / .- Source state '*' = current best, '+' = combined, '-' = not combined,
| /             'x' = may be in error, '~' = too variable, '?' = unusable.
||                                                 .- xxxx [ yyyy ] +/- zzzz
||      Reachability register (octal) -.           |  xxxx = adjusted offset,
||      Log2(Polling interval) --.      |          |  yyyy = measured offset,
||                                \     |          |  zzzz = estimated error.
||                                 |    |           \
MS Name/IP address         Stratum Poll Reach LastRx Last sample
===============================================================================
^- prod-ntp-3.ntp1.ps5.cano>     2   6    17    64    +19ms[  +19ms] +/-  107ms
^- prod-ntp-5.ntp4.ps5.cano>     2   6    17    65  +1185us[+1733us] +/-   89ms
^? alphyn.canonical.com          2   6    17    63   -619us[  -72us] +/-   68ms
^- prod-ntp-4.ntp1.ps5.cano>     2   6    37     1  +5549us[+5549us] +/-   92ms
^- 0.co.ntp.edgeuno.com          2   6    37     0   +346us[ +346us] +/-  156ms
^- Static-Somos-38-19-83-11>     3   6    37     1  -2902us[-2902us] +/-   25ms
^- 1.co.ntp.edgeuno.com          2   6    17    42  -1835us[-1287us] +/-  130ms
^- 0.cl.ntp.edgeuno.com          2   6    37     2  -5726us[-5726us] +/-  174ms
^* 192.168.1.34                  3   6    37     4    +52us[ +599us] +/-   71ms
ubuntu@nutresa-cliente:~$
```

- Configurar NFSv4 (archivos compartidos)

Guia



Instalamos el servidor NFS en el server

```
ubuntu@nutresa-server:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@nutresa-server:~$ sudo apt install -y nfs-kernel-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nfs-kernel-server is already the newest version (1:2.6.4-3ubuntu5.1).
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
ubuntu@nutresa-server:~$
```

Se activa y se comprueba

```
ubuntu@nutresa-server:~$ sudo systemctl enable --now nfs-kernel-server
Synchronizing state of nfs-kernel-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nfs-kernel-server
ubuntu@nutresa-server:~$ sudo systemctl status nfs-kernel-server
● nfs-server.service - NFS server and services
     Loaded: loaded (/usr/lib/systemd/system/nfs-server.service; enabled; preset: enabled)
     Active: active (exited) since Tue 2026-11-10 16:32:49 -05; 1h 41min ago
   Main PID: 5730 (code=exited, status=0/SUCCESS)
        CPU: 10ms

Nov 10 16:32:49 nutresa-server systemd[1]: Starting nfs-server.service - NFS server and services...
Nov 10 16:32:49 nutresa-server exportfs[5729]: exportfs: can't open /etc/exports for reading
Nov 10 16:32:49 nutresa-server systemd[1]: Finished nfs-server.service - NFS server and services.
ubuntu@nutresa-server:~$ sudo exportfs -v
ubuntu@nutresa-server:~$
```

Se crea el grupo corporativo:Vamos a utilizar el grupo nutresa con GID 2000. Esto es importante porque NFS utiliza identificadores numéricos para gestionar los permisos.

```
ubuntu@nutresa-server:~$ sudo groupadd -g 2000 nutresa
ubuntu@nutresa-server:~$ sudo mkdir -p /srv/nfs/nutresa
ubuntu@nutresa-server:~$ sudo chown root:nutresa /srv/nfs/nutresa
sudo chmod 2770 /srv/nfs/nutresa
ubuntu@nutresa-server:~$ ls -ld /srv/nfs/nutresa
drwxrws--- 2 root nutresa 4096 Nov 10 18:17 /srv/nfs/nutresa
ubuntu@nutresa-server:~$
```

Se creó un directorio compartido, se asignaron propietario y permisos, se comprobó. El número 2 de 2770 activa el bit setgid, haciendo que los archivos y directorios nuevos hereden el grupo nutresa.
Configurar exportación NFS

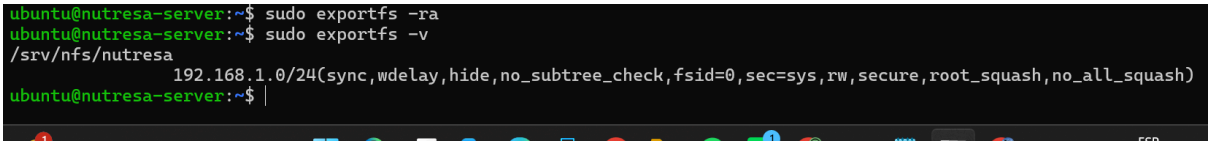
```
ubuntu@nutresa-server: ~
  GNU nano 7.2                                    /etc/exports
# /etc/exports: the access control list for filesystems which may be exported
#               to NFS clients.  See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes       hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_subtree_check)
#
# Example for NFSv4:
# /srv/nfs4        gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes  gss/krb5i(rw,sync,no_subtree_check)
#
/srv/nfs/nutresa 192.168.1.0/24(rw,sync,no_subtree_check,root_squash,fsid=0)
```

que significa:

Opción	Función

rw	Permite lectura y escritura
sync	Confirma escrituras de forma sincronizada
no_subtree_check	Evita comprobaciones adicionales de subdirectorios
root_squash	Evita que el root remoto tenga privilegios de root en el servidor
fsid=0	Define la raíz de exportación NFSv4

Se aplico una configuración y se comprobo con la salida de la carpeta y direccion red en uso autorizada

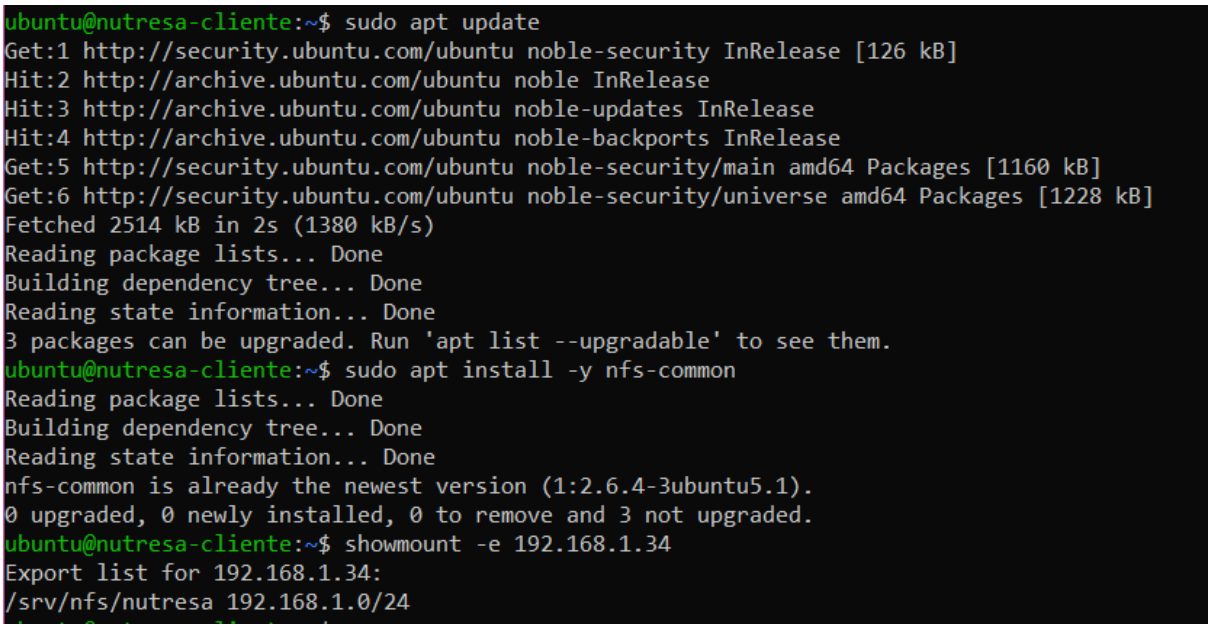


Se comprueba el puerto NFS



Cliente

Actualizacion e Instalacion las herramientas NFS



Crear el grupo con el mismo GID

```
ubuntu@nutresa-cliente:~$ sudo groupadd -g 2000 nutresa
ubuntu@nutresa-cliente:~$ sudo usermod -aG nutresa ubuntu
ubuntu@nutresa-cliente:~$ getent group nutresa
nutresa:x:2000:ubuntu
ubuntu@nutresa-cliente:~$ newgrp nutresa
ubuntu@nutresa-cliente:~$ id
uid=1000(ubuntu) gid=2000(nutresa) groups=2000(nutresa),4(adm),24(cdrom),27(sudo),30(dip),105(lxd),1000(
ubuntu)
```

Creación del punto de montaje

```
ubuntu@nutresa-cliente:~$ sudo mkdir -p /mnt/nutresa
ubuntu@nutresa-cliente:~$ sudo mount -t nfs4 192.168.1.34:/ /mnt/nutresa
ubuntu@nutresa-cliente:~$ df -hT /mnt/nutresa
Filesystem     Type  Size  Used Avail Use% Mounted on
192.168.1.34:/ nfs4   38G  2.4G   36G   7% /mnt/nutresa
ubuntu@nutresa-cliente:~$ mount | grep nutresa
192.168.1.34:/ on /mnt/nutresa type nfs4 (rw,relatime,vers=4.2,rsize=524288,wsize=524288,namlen=255,hard,fat
al_neterrors=none,proto=tcp,timeo=600,retrans=2,sec=sys,clientaddr=192.168.1.35,local_lock=none,addr=192.168
.1.34)
ubuntu@nutresa-cliente:~$
```

Probar escritura y lectura desde el cliente

```
ubuntu@nutresa-cliente:~$ touch /mnt/nutresa/prueba_cliente.txt
ubuntu@nutresa-cliente:~$ echo "Prueba NFS desde el cliente Nutresa" > /mnt/nutresa/prueba_cliente.txt
ubuntu@nutresa-cliente:~$ cat /mnt/nutresa/prueba_cliente.txt
Prueba NFS desde el cliente Nutresa
ubuntu@nutresa-cliente:~$
```

Servidor

Cuando se intentó revisar el archivo desde cliente a servidor se mostro un error, aunque es txt si llego correctamente y tuviera los permisos adecuados

```
ubuntu@nutresa-server:~$ ls -l /srv/nfs/nutresa/
ls: cannot open directory '/srv/nfs/nutresa/': Permission denied
ubuntu@nutresa-server:~$ ls -ld /srv/nfs/nutresa
drwxrws--- 2 root nutresa 4096 Nov 10 19:08 /srv/nfs/nutresa
ubuntu@nutresa-server:~$ sudo ls -l /srv/nfs/nutresa/
total 4
-rw-rw-r-- 1 ubuntu nutresa 36 Nov 10 19:08 prueba_cliente.txt
ubuntu@nutresa-server:~$
```

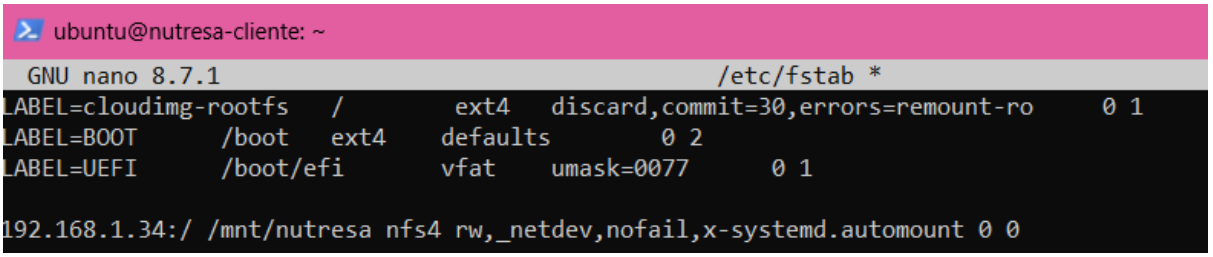
El problema viene porque ubuntu no entra en el grupo ya que solo esta root. Entonces lo que se hizo fue agregar a ubuntu y confirmar que apareciera dentro del grupo. Luego se accedió satisfactoriamente

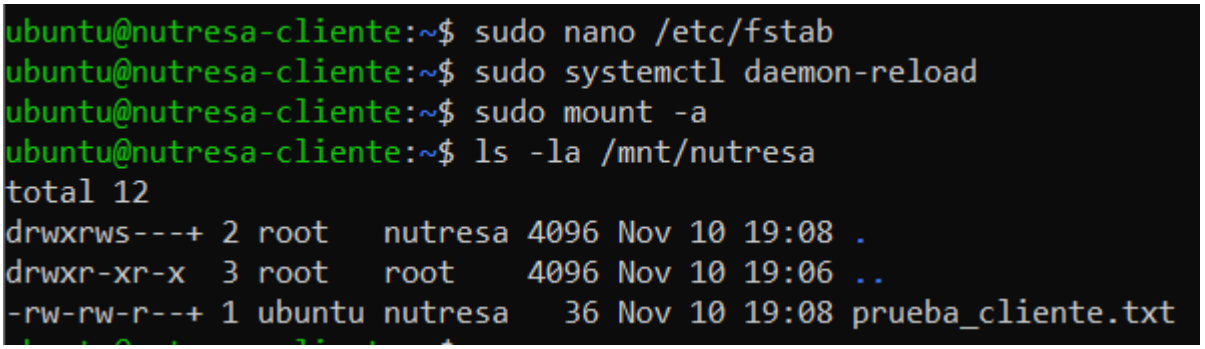
```
ubuntu@nutresa-server:~$ sudo usermod -aG nutresa ubuntu
ubuntu@nutresa-server:~$ newgrp nutresa
ubuntu@nutresa-server:~$ groups
ls -l /srv/nfs/nutresa/
nutresa adm cdrom sudo dip lxd ubuntu
total 4
-rw-rw-r-- 1 ubuntu nutresa 36 Nov 10 19:08 prueba_cliente.txt
ubuntu@nutresa-server:~$ ls -l /srv/nfs/nutresa/
total 4
-rw-rw-r-- 1 ubuntu nutresa 36 Nov 10 19:08 prueba_cliente.txt
ubuntu@nutresa-server:~$ cat /srv/nfs/nutresa/prueba_cliente.txt
Prueba NFS desde el cliente Nutresa
ubuntu@nutresa-server:~$
```

Hacer permanente el montaje NFS

Cliente

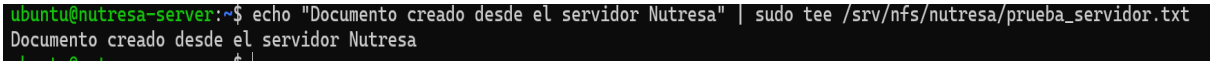
Hasta ahora el montaje funciona, pero si el cliente se reinicia, podría perderse. Vamos a configurarlo para que se monte automáticamente.



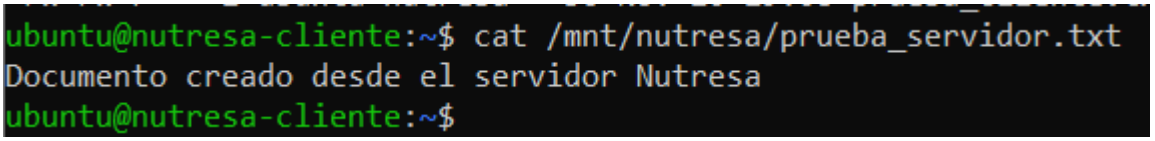


Servidor

Se crea un archivo desde server, el cual se va a poder visualizar desde el cliente

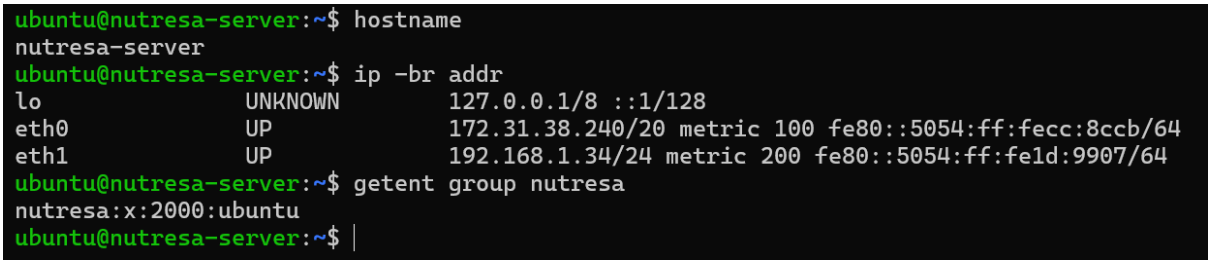


desde el cliente:



IMPLEMENTACIÓN NIS

Preparación del servidor: Se confirmó el nombre del servidor, la ip, y el grupo, todo está correctamente configurado



Comprobar e instalar los paquetes NIS. ypserv y ypinit están disponibles, y que ypinit se invoca con la ruta completa /usr/lib/yp/ypinit.

```
ubuntu@nutresa-server:~$ sudo apt install -y nis
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nis is already the newest version (4.5).
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
ubuntu@nutresa-server:~$ dpkg -l | grep -E 'nis|ypserv|rpcbind'
ii  iptables                      1.8.10-3ubuntu2                         amd64        administration tools for packet fil
tering and NAT
ii  landscape-common              24.02-0ubuntu5.7                        amd64        Landscape administration system cli
ent - Common files
ii  libgssapi-krb5-2:amd64        1.20.1-6ubuntu2.10                      amd64        MIT Kerberos runtime libraries - kr
b5 GSS-API Mechanism
ii  libnss-nis:amd64              3.1-0ubuntu7                            amd64        NSS module for using NIS as a namin
g service
ii  libopeniscsiusr               2.1.9-3ubuntu5.5                        amd64        iSCSI userspace library
ii  libunistring5:amd64           1.1-2build1.1                           amd64        Unicode string library for C
ii  nis                           4.5                                     all          Network Information Service (NIS)
ii  passwd                        1:4.13+dfsg1-4ubuntu3.2                 amd64        change and administer password and
group data
ii  polkitd                       124-2ubuntu1.24.04.4                    amd64        framework for managing administrati
ve policies and privileges
ii  rpcbind                       1.2.6-7ubuntu2                          amd64        converts RPC program numbers into u
niversal addresses
ii  ypserv                        4.2-2build2                             amd64        Server daemon for working with Netw
ork Information System (NIS)
ubuntu@nutresa-server:~$ command -v ypserv
/usr/sbin/ypserv
ubuntu@nutresa-server:~$ command -v ypinit
ubuntu@nutresa-server:~$
```

Instalar explícitamente el servidor NIS: paquetes instalados con estado ii.

```
ubuntu@nutresa-server:~$ sudo apt install -y ypserv rpcbind
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ypserv is already the newest version (4.2-2build2).
ypserv set to manually installed.
rpcbind is already the newest version (1.2.6-7ubuntu2).
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
ubuntu@nutresa-server:~$ dpkg -l | grep -E 'nis|ypserv|yp-tools|rpcbind'
ii  iptables                      1.8.10-3ubuntu2                         amd64        administration tools for packet fil
tering and NAT
ii  landscape-common              24.02-0ubuntu5.7                        amd64        Landscape administration system cli
ent - Common files
ii  libgssapi-krb5-2:amd64        1.20.1-6ubuntu2.10                      amd64        MIT Kerberos runtime libraries - kr
b5 GSS-API Mechanism
ii  libnss-nis:amd64              3.1-0ubuntu7                            amd64        NSS module for using NIS as a namin
g service
ii  libopeniscsiusr               2.1.9-3ubuntu5.5                        amd64        iSCSI userspace library
ii  libunistring5:amd64           1.1-2build1.1                           amd64        Unicode string library for C
ii  nis                           4.5                                     all          Network Information Service (NIS)
ii  passwd                        1:4.13+dfsg1-4ubuntu3.2                 amd64        change and administer password and
group data
ii  polkitd                       124-2ubuntu1.24.04.4                    amd64        framework for managing administrati
ve policies and privileges
ii  rpcbind                       1.2.6-7ubuntu2                          amd64        converts RPC program numbers into u
niversal addresses
ii  yp-tools                      4.2.3-4build1                           amd64        Tools for working with Network Info
rmation System (NIS)
ii  ypserv                        4.2-2build2                             amd64        Server daemon for working with Netw
ork Information System (NIS)
ubuntu@nutresa-server:~$ command -v ypserv
command -v ypinit
/usr/sbin/ypserv
```

Establecer el dominio NIS :

```
ubuntu@nutresa-server:~$ echo "nutresa.local" | sudo tee /etc/defaultdomain
nutresa.local
ubuntu@nutresa-server:~$ sudo domainname nutresa.local
ubuntu@nutresa-server:~$ domainname
nutresa.local
ubuntu@nutresa-server:~$
```

Configurar los puertos fijos:

```
ubuntu@nutresa-server:~$ sudo nano /etc/default/nis
ubuntu@nutresa-server:~$ sudo systemctl restart ypserv yppasswdd
ubuntu@nutresa-server:~$ rpcinfo -p | grep -E 'ypserv|yppasswdd'
sudo ss -lntup | grep -E '1011|1012'
    100004    2   udp   1011  ypserv
    100004    1   udp   1011  ypserv
    100004    2   tcp   1011  ypserv
    100004    1   tcp   1011  ypserv
    100009    1   udp   1012  yppasswdd
    100009    1   tcp   1012  yppasswdd
udp   UNCONN 0      0               0.0.0.0:1011       0.0.0.0:*    users:(("ypserv",pid=9570,fd=5))
udp   UNCONN 0      0               0.0.0.0:1012       0.0.0.0:*    users:(("rpc.yppasswdd",pid=9575,fd=5))
udp   UNCONN 0      0                  [::]:1011          [::]:*    users:(("ypserv",pid=9570,fd=7))
udp   UNCONN 0      0                  [::]:1012          [::]:*    users:(("rpc.yppasswdd",pid=9575,fd=7))
tcp   LISTEN 0      4096            0.0.0.0:1012       0.0.0.0:*    users:(("rpc.yppasswdd",pid=9575,fd=6))
tcp   LISTEN 0      4096            0.0.0.0:1011       0.0.0.0:*    users:(("ypserv",pid=9570,fd=6))
tcp   LISTEN 0      4096               [::]:1012          [::]:*    users:(("rpc.yppasswdd",pid=9575,fd=8))
tcp   LISTEN 0      4096               [::]:1011          [::]:*    users:(("ypserv",pid=9570,fd=8))
ubuntu@nutresa-server:~$
```

```
ubuntu@nutresa-server: ~
  GNU nano 7.2                                  /etc/default/nis
#
# /etc/defaults/nis     Optional configuration settings for the NIS programs.
#

# Are we a NIS server and if so what kind (values: false, slave, master)?
NISSERVER=master

# Are we a NIS client?
NISCLIENT=false

# Location of the master NIS password file (for yppasswdd).
# If you change this make sure it matches with /var/yp/Makefile.
YPPWDDIR=/etc

# Do we allow the user to use ypchsh and/or ypchfn ? The YPCHANGEOK
# fields are passed with -e to yppasswdd, see it's manpage.
# Possible values: "chsh", "chfn", "chsh,chfn"
YPCHANGEOK=chsh

# NIS master server.  If this is configured on a slave server then ypinit
# will be run each time NIS is started.
NISMASTER=

# Additional options to be given to ypserv when it is started.
YPSERVARGS="-p 1011"

# Additional options to be given to ypbind when it is started.
YPBINDARGS=

                                [ Read 35 lines ]
^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

Restringir el acceso NIS a la red local : Hay que **borrar** las dos líneas que dan acceso a todo el mundo (0.0.0.0 0.0.0.0 y ::/0) y agregar nuestra subred

```
ubuntu@nutresa-server: ~
  GNU nano 7.2                                  /etc/ypserv.secure
#
# securenets    This file defines the access rights to your NIS server
#               for NIS clients. This file contains netmask/network
#               pairs. A clients IP address needs to match with at least
#               one of those.
#
#               One can use the word "host" instead of a netmask of
#               255.255.255.255. Only IP addresses are allowed in this
#               file, not hostnames.
#
# Always allow access for localhost, IPv4 and IPv6
255.0.0.0       127.0.0.0
255.255.255.0   192.168.1.0
```

Crear un usuario de prueba centralizado :Se crea un usuario exclusivo para probar NIS

```
ubuntu@nutresa-server:~$ getent group 2000
nutresa:x:2000:ubuntu
ubuntu@nutresa-server:~$ sudo useradd -m -u 2101 -g 2000 -s /bin/bash nisprueba
ubuntu@nutresa-server:~$ sudo passwd nisprueba
New password:
Retype new password:
passwd: password updated successfully
ubuntu@nutresa-server:~$ id nisprueba
uid=2101(nisprueba) gid=2000(nutresa) groups=2000(nutresa)
ubuntu@nutresa-server:~$
```

Inicializar la base de datos NIS:
dominio NIS: nutresa.local
La primera vez no mostraba el usuario en el servidor NIS, esto debido a que ypbind dice “No NIS server and no -broadcast option specified”. Es decir, /etc/yp.conf no tiene ninguna línea con el servidor. Probablemente nunca se guardó la línea que te indiqué antes, o el archivo solo tiene comentarios.

```
ubuntu@nutresa-server:~$ sudo /usr/lib/yp/makedbm -u /var/yp/nutresa.local/passwd.byname | grep nisprueba
nisprueba       nisprueba:x:2101:2000::/home/nisprueba:/bin/bash
ubuntu@nutresa-server:~$ sudo systemctl restart rpcbind
ubuntu@nutresa-server:~$ sudo systemctl restart ypserv yppasswdd
ubuntu@nutresa-server:~$ sudo systemctl restart ypbind
Job for ypbind.service failed because the control process exited with error code.
See "systemctl status ypbind.service" and "journalctl -xeu ypbind.service" for details.
ubuntu@nutresa-server:~$ journalctl -xeu ypbind.service -n 30 --no-pager
Nov 10 20:24:33 nutresa-server systemd[1]: Starting ypbind.service - NIS Binding Service...
    Subject: A start job for unit ypbind.service has begun execution
    Defined-By: systemd
    Support: http://www.ubuntu.com/support

    A start job for unit ypbind.service has begun execution.

    The job identifier is 10192.
Nov 10 20:24:33 nutresa-server ypbind[10048]: No NIS server and no -broadcast option specified.
Nov 10 20:24:33 nutresa-server ypbind[10048]: Add a NIS server to the /etc/yp.conf configuration file,
Nov 10 20:24:33 nutresa-server ypbind[10048]: or start ypbind with the -broadcast option.
Nov 10 20:24:33 nutresa-server systemd[1]: ypbind.service: Control process exited, code=exited, status=1/FAILURE
    Subject: Unit process exited
    Defined-By: systemd
    Support: http://www.ubuntu.com/support

    An ExecStart= process belonging to unit ypbind.service has exited.

    The process' exit code is 'exited' and its exit status is 1.
Nov 10 20:24:33 nutresa-server systemd[1]: ypbind.service: Failed with result 'exit-code'.
    Subject: Unit failed
    Defined-By: systemd
    Support: http://www.ubuntu.com/support

    The unit ypbind.service has entered the 'failed' state with result 'exit-code'.
Nov 10 20:24:33 nutresa-server systemd[1]: Failed to start ypbind.service - NIS Binding Service.
    Subject: A start job for unit ypbind.service has failed
    Defined-By: systemd
    Support: http://www.ubuntu.com/support

    A start job for unit ypbind.service has finished with a failure.

    The job identifier is 10192 and the job result is failed.
ubuntu@nutresa-server:~$ grep -v '^#' /etc/yp.conf
```

aca ya se confirmó la creación de los mapas NIS y que el usuario de prueba creado este dentro

```
ubuntu@nutresa-server:~$ sudo make -C /var/yp
make: Entering directory '/var/yp'
gmake[1]: Entering directory '/var/yp/nutresa.local'
gmake[1]: Nothing to be done for 'all'.
gmake[1]: Leaving directory '/var/yp/nutresa.local'
make: Leaving directory '/var/yp'
ubuntu@nutresa-server:~$ ls -l /var/yp/nutresa.local/
total 400
-rw------- 1 root root  16384 Nov 10 20:20 group.bygid
-rw------- 1 root root  16384 Nov 10 20:20 group.byname
-rw------- 1 root root  16384 Nov 10 20:20 hosts.byaddr
-rw------- 1 root root  16384 Nov 10 20:20 hosts.byname
-rw------- 1 root root  16384 Nov 10 20:20 netgroup
-rw------- 1 root root  16384 Nov 10 20:20 netgroup.byhost
-rw------- 1 root root  16384 Nov 10 20:20 netgroup.byuser
-rw------- 1 root root  16384 Nov 10 20:20 netid.byname
-rw------- 1 root root  16384 Nov 10 20:20 passwd.byname
-rw------- 1 root root  16384 Nov 10 20:20 passwd.byuid
-rw------- 1 root root  16384 Nov 10 20:20 protocols.byname
-rw------- 1 root root  16384 Nov 10 20:20 protocols.bynumber
-rw------- 1 root root  16384 Nov 10 20:20 rpc.byname
-rw------- 1 root root  16384 Nov 10 20:20 rpc.bynumber
-rw------- 1 root root  45056 Nov 10 20:20 services.byname
-rw------- 1 root root 102400 Nov 10 20:20 services.byservicename
-rw------- 1 root root  16384 Nov 10 20:20 shadow.byname
-rw------- 1 root root  16384 Nov 10 20:20 ypservers
ubuntu@nutresa-server:~$ sudo ypmatch nisprueba passwd.byname
nisprueba:x:2101:2000::/home/nisprueba:/bin/bash
ubuntu@nutresa-server:~$
```

Activar los servicios NIS :

```
ubuntu@nutresa-server:~$ sudo systemctl enable rpcbind
sudo systemctl restart rpcbind
sudo systemctl restart ypserv
Synchronizing state of rpcbind.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable rpcbind
ubuntu@nutresa-server:~$ sudo systemctl status ypserv
● ypserv.service - NIS/YP (Network Information Service) Server
     Loaded: loaded (/usr/lib/systemd/system/ypserv.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-11-10 20:34:21 -05; 14s ago
    Process: 10284 ExecStartPre=/bin/domainname -F /etc/defaultdomain (code=exited, status=0/SUCCESS)
    Process: 10289 ExecStart=/usr/sbin/ypserv $YPSERVARGS (code=exited, status=0/SUCCESS)
   Main PID: 10291 (ypserv)
      Tasks: 1 (limit: 4614)
     Memory: 644.0K (peak: 1.5M)
        CPU: 12ms
     CGroup: /system.slice/ypserv.service
             └─10291 /usr/sbin/ypserv -p 1011

Nov 10 20:34:21 nutresa-server systemd[1]: Starting ypserv.service - NIS/YP (Network Information Service) Server...
Nov 10 20:34:21 nutresa-server systemd[1]: ypserv.service: Can't open PID file /run/ypserv.pid (yet?) after start: No such file or directory
Nov 10 20:34:21 nutresa-server systemd[1]: Started ypserv.service - NIS/YP (Network Information Service) Server.
ubuntu@nutresa-server:~$ sudo systemctl status rpcbind
● rpcbind.service - RPC bind portmap service
     Loaded: loaded (/usr/lib/systemd/system/rpcbind.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-11-10 20:34:21 -05; 33s ago
TriggeredBy: ● rpcbind.socket
       Docs: man:rpcbind(8)
   Main PID: 10271 (rpcbind)
      Tasks: 1 (limit: 4614)
     Memory: 616.0K (peak: 876.0K)
        CPU: 7ms
     CGroup: /system.slice/rpcbind.service
             └─10271 /sbin/rpcbind -f -w

Nov 10 20:34:21 nutresa-server systemd[1]: Starting rpcbind.service - RPC bind portmap service...
Nov 10 20:34:21 nutresa-server systemd[1]: Started rpcbind.service - RPC bind portmap service.
ubuntu@nutresa-server:~$ sudo systemctl enable --now yppasswdd
Synchronizing state of yppasswdd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable yppasswdd
ubuntu@nutresa-server:~$
```

Abrir los puertos NIS : Por ahora no vamos a activar el firewall ni el UFW, las reglas quedan preparadas.

```
ubuntu@nutresa-server:~$ sudo ufw allow from 192.168.1.0/24 to any port 111 proto tcp
sudo ufw allow from 192.168.1.0/24 to any port 111 proto udp
sudo ufw allow from 192.168.1.0/24 to any port 1011 proto tcp
sudo ufw allow from 192.168.1.0/24 to any port 1011 proto udp
sudo ufw allow from 192.168.1.0/24 to any port 1012 proto tcp
sudo ufw allow from 192.168.1.0/24 to any port 1012 proto udp
Rules updated
Rules updated
Rules updated
Rules updated
Rules updated
Rules updated
ubuntu@nutresa-server:~$ sudo ufw status
Status: inactive
ubuntu@nutresa-server:~$
```

Configurar el cliente NIS **Cliente**:

```
ubuntu@nutresa-cliente:~$ sudo apt install -y nis rpcbind
rpcbind is already the newest version (1.2.7-1build2).
rpcbind set to manually installed.
Installing:
  nis

Installing dependencies:
  libnsl2  libnss-nis  make  nscd  yp-tools  ypbind-mt  ypserv

Suggested packages:
  make-doc  krb5-kdc

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 58
  Download size: 552 kB
  Space needed: 1744 kB / 1651 MB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libnsl2 amd64 1.3.0-3build4 [41.8 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 make amd64 4.4.1-3 [197 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/universe amd64 ypserv amd64 4.2-3 [111 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libnss-nis amd64 3.1-5build1 [26.6 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 nscd amd64 2.43-2ubuntu2.4 [74.0 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute/universe amd64 ypbind-mt amd64 2.7.2-3build2 [35.9 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/universe amd64 yp-tools amd64 4.2.3-6 [52.5 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute/universe amd64 nis all 4.5build1 [13.4 kB]
Fetched 552 kB in 1s (645 kB/s)
Selecting previously unselected package libnsl2:amd64.
(Reading database ... 91629 files and directories currently installed.)
Preparing to unpack .../0-libnsl2_1.3.0-3build4_amd64.deb ...
Unpacking libnsl2:amd64 (1.3.0-3build4) ...
Selecting previously unselected package make.
Preparing to unpack .../1-make_4.4.1-3_amd64.deb ...
Unpacking make (4.4.1-3) ...
Selecting previously unselected package ypserv.
Preparing to unpack .../2-ypserv_4.2-3_amd64.deb ...
Unpacking ypserv (4.2-3) ...
Selecting previously unselected package libnss-nis:amd64.
Preparing to unpack .../3-libnss-nis_3.1-5build1_amd64.deb ...
Unpacking libnss-nis:amd64 (3.1-5build1) ...
Selecting previously unselected package nscd.
Preparing to unpack .../4-nscd_2.43-2ubuntu2.4_amd64.deb ...
Unpacking nscd (2.43-2ubuntu2.4) ...
Selecting previously unselected package ypbind-mt.
Preparing to unpack .../5-ypbind-mt_2.7.2-3build2_amd64.deb ...
Unpacking ypbind-mt (2.7.2-3build2) ...
Selecting previously unselected package yp-tools.
Preparing to unpack .../6-yp-tools_4.2.3-6_amd64.deb ...
Unpacking yp-tools (4.2.3-6) ...
Selecting previously unselected package nis.
```

```
ubuntu@nutresa-cliente:~$ sudo domainname nutresa.local
ubuntu@nutresa-cliente:~$ echo "nutresa.local" | sudo tee /etc/defaultdomain
sudo domainname nutresa.local
nutresa.local
ubuntu@nutresa-cliente:~$ cat /etc/defaultdomain
domainname
nutresa.local
nutresa.local
ubuntu@nutresa-cliente:~$ sudo nano /etc/default/nis
ubuntu@nutresa-cliente:~$ echo "domain nutresa.local server 192.168.1.34" | sudo tee -a /etc/yp.conf
domain nutresa.local server 192.168.1.34
ubuntu@nutresa-cliente:~$ sudo nano /etc/nsswitch.conf
ubuntu@nutresa-cliente:~$ sudo nano /etc/nsswitch.conf
ubuntu@nutresa-cliente:~$ sudo systemctl enable --now rpcbind ypbind
sudo systemctl restart ypbind
systemctl is-active ypbind
ypwhich
ypcat passwd
ypmatch nisprueba passwd.byname
getent passwd nisprueba
Synchronizing state of rpcbind.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable rpcbind
Synchronizing state of ypbind.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ypbind
Created symlink '/etc/systemd/system/multi-user.target.wants/ypbind.service' → '/usr/lib/systemd/system/ypbind.s
ervice'.
active
192.168.1.34
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
nisprueba:x:2101:2000::/home/nisprueba:/bin/bash
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
nisprueba:x:2101:2000::/home/nisprueba:/bin/bash
nisprueba:x:2101:2000::/home/nisprueba:/bin/bash
ubuntu@nutresa-cliente:~$ sudo domainname nutresa.local
ubuntu@nutresa-cliente:~$ echo "nutresa.local" | sudo tee /etc/defaultdomain
nutresa.local
ubuntu@nutresa-cliente:~$ sudo nano /etc/yp.conf
ubuntu@nutresa-cliente:~$ sudo systemctl enable --now rpcbind
Synchronizing state of rpcbind.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable rpcbind
ubuntu@nutresa-cliente:~$ sudo systemctl enable --now ypbind
Synchronizing state of ypbind.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ypbind
ubuntu@nutresa-cliente:~$ systemctl list-unit-files | grep -E 'ypbind|nis'
ypbind.service                            enabled         enabled
ubuntu@nutresa-cliente:~$
```

Configurar la consulta de usuarios NIS:

```
ubuntu@nutresa-cliente:~$ ypwhich
192.168.1.34
ubuntu@nutresa-cliente:~$ getent passwd nisprueba
nisprueba:x:2101:2000::/home/nisprueba:/bin/bash
ubuntu@nutresa-cliente:~$ getent group nutresa
nutresa:x:2000:ubuntu
ubuntu@nutresa-cliente:~$
```

IMPLEMENTACIÓN SAMBA

Instalación de Samba: Se actualiza con “update” y luego se instala samba, se verifica la versión y el estatus

```
ubuntu@nutresa-server:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@nutresa-server:~$ sudo apt install -y samba smbclient
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
samba is already the newest version (2:4.19.5+dfsg-4ubuntu9.8).
Suggested packages:
  cifs-utils heimdal-clients
The following NEW packages will be installed:
  libsmbclient0 smbclient
0 upgraded, 2 newly installed, 0 to remove and 3 not upgraded.
Need to get 533 kB of archives.
After this operation, 2376 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libsmbclient0 amd64 2:4.19.5+dfsg-4ubuntu9.8 [62.4 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 smbclient amd64 2:4.19.5+dfsg-4ubuntu9.8 [470 kB]
Fetched 533 kB in 2s (236 kB/s)
Selecting previously unselected package libsmbclient0:amd64.
(Reading database ... 113486 files and directories currently installed.)
Preparing to unpack .../libsmbclient0_2%3a4.19.5+dfsg-4ubuntu9.8_amd64.deb ...
Unpacking libsmbclient0:amd64 (2:4.19.5+dfsg-4ubuntu9.8) ...
Selecting previously unselected package smbclient.
Preparing to unpack .../smbclient_2%3a4.19.5+dfsg-4ubuntu9.8_amd64.deb ...
Unpacking smbclient (2:4.19.5+dfsg-4ubuntu9.8) ...
Setting up libsmbclient0:amd64 (2:4.19.5+dfsg-4ubuntu9.8) ...
Setting up smbclient (2:4.19.5+dfsg-4ubuntu9.8) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.9) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.
```

```
ubuntu@nutresa-server:~$ smbd --version
Version 4.19.5-Ubuntu
ubuntu@nutresa-server:~$ sudo systemctl enable --now smbd
Synchronizing state of smbd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable smbd
ubuntu@nutresa-server:~$ sudo systemctl status smbd
● smbd.service - Samba SMB Daemon
     Loaded: loaded (/usr/lib/systemd/system/smbd.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-11-10 16:32:56 -05; 4h 36min ago
       Docs: man:smbd(8)
             man:samba(7)
             man:smb.conf(5)
   Main PID: 6751 (smbd)
     Status: "smbd: ready to serve connections..."
      Tasks: 3 (limit: 4614)
     Memory: 10.2M (peak: 10.9M)
        CPU: 221ms
     CGroup: /system.slice/smbd.service
             ├─6751 /usr/sbin/smbd --foreground --no-process-group
             ├─6755 "smbd: notifyd" .
             └─6756 "smbd: cleanupd "

Nov 10 16:32:56 nutresa-server systemd[1]: Starting smbd.service - Samba SMB Daemon...
Nov 10 16:32:56 nutresa-server (smbd)[6751]: smbd.service: Referenced but unset environment variable evaluates to an empty string: SMBDOPTIONS
Nov 10 16:32:56 nutresa-server systemd[1]: Started smbd.service - Samba SMB Daemon.
ubuntu@nutresa-server:~$
```

Crear el grupo de usuarios Samba : Se crea el usuario con su respectiva contraseña

```
ubuntu@nutresa-server:~$ sudo groupadd sambausers
ubuntu@nutresa-server:~$ sudo useradd -M -s /usr/sbin/nologin sambanutresa
ubuntu@nutresa-server:~$ sudo usermod -aG sambausers sambanutresa
ubuntu@nutresa-server:~$ sudo smbpasswd -a sambanutresa
New SMB password:
Retype new SMB password:
Added user sambanutresa.
ubuntu@nutresa-server:~$ sudo smbpasswd -e sambanutresa
Enabled user sambanutresa.
ubuntu@nutresa-server:~$ sudo pdbedit -L
sambanutresa:2102:
ubuntu@nutresa-server:~$
```

Crear la carpeta corporativa y asignar permisos, luego se comprueba con ls y por último se crea un archivo de prueba:

```
ubuntu@nutresa-server:~$ sudo mkdir -p /srv/samba/corporativo
ubuntu@nutresa-server:~$ sudo chown root:sambausers /srv/samba/corporativo
sudo chmod 2770 /srv/samba/corporativo
ubuntu@nutresa-server:~$ ls -ld /srv/samba/corporativo
drwxrws--- 2 root sambausers 4096 Nov 10 21:17 /srv/samba/corporativo
ubuntu@nutresa-server:~$ echo "Documentacion corporativa Grupo Nutresa" | sudo tee /srv/samba/corporativo/README.txt
Documentacion corporativa Grupo Nutresa
ubuntu@nutresa-server:~$
```

Configurar el recurso compartido:

```
ubuntu@nutresa-server: ~
 GNU nano 7.2                                  /etc/samba/smb.conf
;   path = /home/samba/profiles
;   guest ok = no
;   browseable = no
;   create mask = 0600
;   directory mask = 0700

[printers]
   comment = All Printers
   browseable = no
   path = /var/tmp
   printable = yes
   guest ok = no
   read only = yes
   create mask = 0700

# Windows clients look for this share name as a source of downloadable
# printer drivers
[print$]
   comment = Printer Drivers
   path = /var/lib/samba/printers
   browseable = yes
   read only = yes
   guest ok = no
# Uncomment to allow remote administration of Windows print drivers.
# You may need to replace 'lpadmin' with the name of the group your
# admin users are members of.
# Please note that you also need to set appropriate Unix permissions
# to the drivers directory for these users to have write rights in it
;   write list = root, @lpadmin

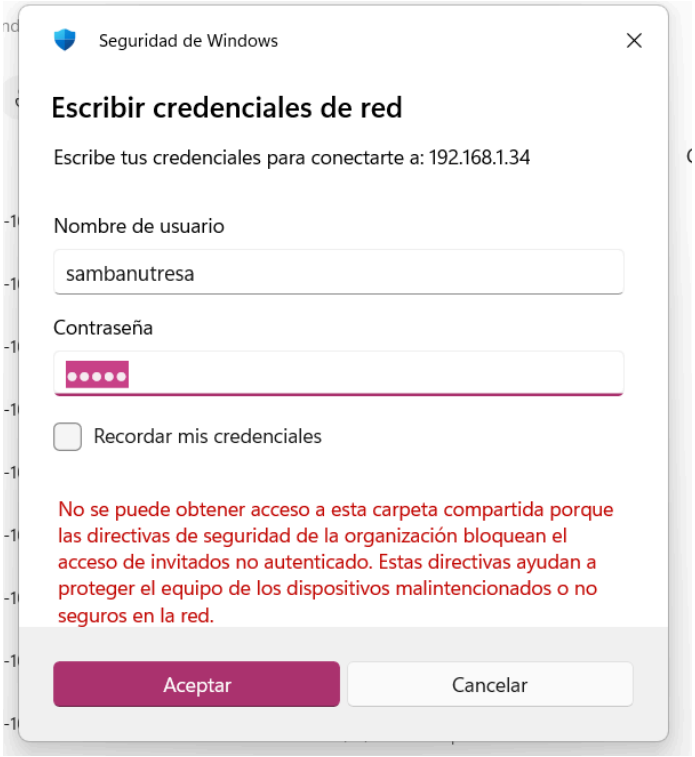
[Corporativo]
   path = /srv/samba/corporativo
   browseable = yes
   read only = no
   guest ok = no
   valid users = @sambausers
   force group = sambausers
   create mask = 0660
   directory mask = 2770
```

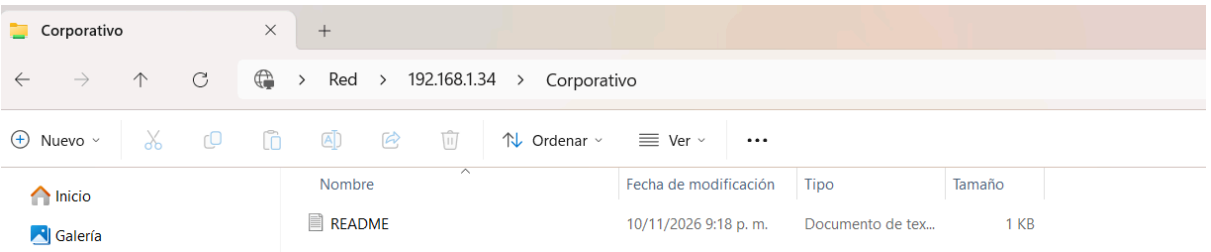
```
        valid users = @sambausers
ubuntu@nutresa-server:~$ sudo systemctl restart smbd
ubuntu@nutresa-server:~$
```

Abrir los puertos Samba:

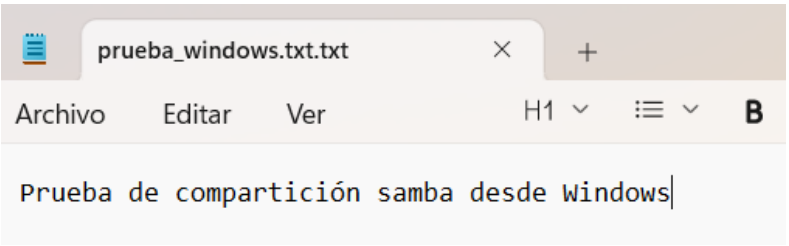
```
ubuntu@nutresa-server:~$ sudo ufw allow from 192.168.1.0/24 to any port 445 proto tcp
sudo ufw allow from 192.168.1.0/24 to any port 139 proto tcp
sudo ufw allow from 192.168.1.0/24 to any port 137 proto udp
sudo ufw allow from 192.168.1.0/24 to any port 138 proto udp
Rules updated
Rules updated
Rules updated
Rules updated
ubuntu@nutresa-server:~$
```

Probar Samba desde Windows **Cliente**:
Se ingresa las credenciales y en Windows se autenticó con sambanutresa y la conexión SMB al recurso está establecida. Eso confirma que el servidor, el usuario, la contraseña y el firewall están bien.





Desde el windows del cliente se crea un archivo txt de prueba.



Luego desde el server se verificó con “sudo” por temas de seguridad y si funciono

```
ubuntu@nutresa-server:~$ ls -l /srv/samba/corporativo/
ls: cannot open directory '/srv/samba/corporativo/': Permission denied
ubuntu@nutresa-server:~$ sudo ls -l /srv/samba/corporativo/
total 8
-rw-r--r-- 1 root         sambausers 40 Nov 10 21:18 README.txt
-rw-rw---- 1 sambanutresa sambausers 43 Nov 10 21:51 prueba_windows.txt.txt
ubuntu@nutresa-server:~$
```

IMPLEMENTACIÓN CUPS

actualización, instalación y confirmación de estado CUPS :

```
ubuntu@nutresa-server:~$ sudo apt update
sudo apt install -y cups printer-driver-cups-pdf
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1444 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1711 kB]
Fetched 3281 kB in 5s (710 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
cups is already the newest version (2.4.7-1.2ubuntu7.14).
Suggested packages:
  system-config-printer
The following NEW packages will be installed:
  printer-driver-cups-pdf
0 upgraded, 1 newly installed, 0 to remove and 3 not upgraded.
Need to get 25.7 kB of archives.
After this operation, 241 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 printer-driver-cups-pdf amd64 3.0.1-14ubuntu0.24.04.1 [25.7 kB]
Fetched 25.7 kB in 1s (28.7 kB/s)
Selecting previously unselected package printer-driver-cups-pdf.
(Reading database ... 113520 files and directories currently installed.)
Preparing to unpack .../printer-driver-cups-pdf_3.0.1-14ubuntu0.24.04.1_amd64.deb ...
Unpacking printer-driver-cups-pdf (3.0.1-14ubuntu0.24.04.1) ...
Setting up printer-driver-cups-pdf (3.0.1-14ubuntu0.24.04.1) ...
Processing triggers for cups (2.4.7-1.2ubuntu7.14) ...
Updating PPD files for cups-pdf ...
PPD for printer PDF updated
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.
```

```
ubuntu@nutresa-server:~$ sudo systemctl enable --now cups
Synchronizing state of cups.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable cups
ubuntu@nutresa-server:~$ sudo systemctl status cups
● cups.service - CUPS Scheduler
     Loaded: loaded (/usr/lib/systemd/system/cups.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-11-10 21:59:13 -05; 31s ago
TriggeredBy: ● cups.socket
             ● cups.path
       Docs: man:cupsd(8)
   Main PID: 12262 (cupsd)
     Status: "Scheduler is running..."
      Tasks: 2 (limit: 4614)
     Memory: 4.4M (peak: 10.3M)
        CPU: 157ms
     CGroup: /system.slice/cups.service
             ├─12262 /usr/sbin/cupsd -l
             └─12293 /usr/lib/cups/notifier/dbus dbus://

Nov 10 21:59:13 nutresa-server systemd[1]: Starting cups.service - CUPS Scheduler...
Nov 10 21:59:13 nutresa-server systemd[1]: Started cups.service - CUPS Scheduler.
ubuntu@nutresa-server:~$
```

Permitir administración desde la red local, para esto primero se configura un nano:

```
# Listen for connections from the local network.
Port 631
Listen /run/cups/cups.sock

# Show shared printers on the local network.
Browsing No
BrowseLocalProtocols dnssd

# Default authentication type, when authenticati
DefaultAuthType Basic

# Web interface setting...
WebInterface Yes

# Timeout after cupsd exits if idle (applied onl
IdleExitTimeout 60


<Location />
  Order allow,deny
  Allow localhost
  Allow 192.168.1.0/24
</Location>

# Restrict access to the admin pages...
<Location /admin>
  AuthType Default
  Require user @SYSTEM
  Order allow,deny
  Allow localhost
  Allow 192.168.1.0/24
</Location>
```

```
ubuntu@nutresa-server:~$ sudo nano /etc/cups/cupsd.conf
ubuntu@nutresa-server:~$ sudo cupsd -t
Printer drivers are deprecated and will stop working in a future version of CUPS. See https://github.com/OpenPrinting/cups/issues/103
"/etc/cups/cups-files.conf" is OK.
"/etc/cups/cupsd.conf" is OK.
ubuntu@nutresa-server:~$ sudo nano /etc/cups/cupsd.conf
ubuntu@nutresa-server:~$ sudo systemctl restart cups
```

Configurar la impresora virtual PDF :

```
ubuntu@nutresa-server:~$ lpinfo -v | grep -i pdf
lpinfo: Forbidden
ubuntu@nutresa-server:~$ lpinfo -m | grep -i pdf
lsb/usr/cupsfilters/Fuji_Xerox-DocuPrint_CM305_df-PDF.ppd Fuji Xerox DocuPrint CM305 df PDF
lsb/usr/cups-pdf/CUPS-PDF_noopt.ppd Generic CUPS-PDF Printer (no options)
lsb/usr/cups-pdf/CUPS-PDF_opt.ppd Generic CUPS-PDF Printer (w/ options)
lsb/usr/cupsfilters/Generic-PDF_Printer-PDF.ppd Generic PDF Printer
lsb/usr/cupsfilters/HP-Color_LaserJet_CM3530_MFP-PDF.ppd HP Color LaserJet CM3530 MFP PDF
lsb/usr/cupsfilters/Ricoh-PDF_Printer-PDF.ppd Ricoh PDF Printer
ubuntu@nutresa-server:~$ sudo lpinfo -v | grep -i pdf
file cups-pdf:/
ubuntu@nutresa-server:~$ sudo lpadmin -p Nutresa-PDF -E -v cups-pdf:/ -m drv:///sample.drv/generic.ppd
lpadmin: Printer drivers are deprecated and will stop working in a future version of CUPS.
ubuntu@nutresa-server:~$ lpstat -p -d
printer Nutresa-PDF is idle.  enabled since Tue Nov 10 22:22:38 2026
printer PDF is idle.  enabled since Tue Nov 10 21:59:13 2026
system default destination: PDF
ubuntu@nutresa-server:~$
```

Abrir el puerto CUPS, ufw todavia no esta activo entonces la regla queda “registrada”:

```
ubuntu@nutresa-server:~$ sudo ufw allow from 192.168.1.0/24 to any port 631 proto tcp
Rules updated
```

Realizar una prueba de impresión : se tuvo un problema con la direccion donde se busco el PDF

```
ubuntu@nutresa-server:~$ echo "Prueba de impresion centralizada - Grupo Nutresa" > ~/prueba_impresion.txt
ubuntu@nutresa-server:~$ lp -d Nutresa-PDF ~/prueba_impresion.txt
request id is Nutresa-PDF-1 (1 file(s))
ubuntu@nutresa-server:~$ lpstat -o
ubuntu@nutresa-server:~$ lpstat -p Nutresa-PDF
printer Nutresa-PDF is idle.  enabled since Tue Nov 10 22:26:25 2026
ubuntu@nutresa-server:~$ sudo find /var/spool/cups-pdf -type f -mmin -5
ubuntu@nutresa-server:~$ lpstat -W completed
sudo tail -n 30 /var/log/cups/error_log
sudo tail -n 30 /var/log/cups/page_log
Nutresa-PDF-1           ubuntu          1024   Tue Nov 10 22:26:25 2026
W [10/Nov/2026:21:59:14 -0500] CreateProfile failed: org.freedesktop.ColorManager.AlreadyExists:profile id \'PDF-Gray..\' already exists
W [10/Nov/2026:21:59:14 -0500] CreateProfile failed: org.freedesktop.ColorManager.AlreadyExists:profile id \'PDF-RGB..\' already exists
W [10/Nov/2026:22:13:32 -0500] Printer drivers are deprecated and will stop working in a future version of CUPS. See https://github.com/OpenPrin
ting/cups/issues/103
E [10/Nov/2026:22:17:28 -0500] [Client 10] Returning HTTP Forbidden for CUPS-Get-Devices (no URI) from localhost
E [10/Nov/2026:22:20:30 -0500] [CGI] cups-brf must be called as root
E [10/Nov/2026:22:20:30 -0500] [cups-deviced] PID 12804 (cups-brf) stopped with status 1!
tail: cannot open '/var/log/cups/page_log' for reading: No such file or directory
ubuntu@nutresa-server:~$ ls -l ~/PDF/
total 12
-rw------- 1 ubuntu ubuntu 8474 Nov 10 22:26 prueba_impresion.pdf
```

Acceder al panel CUPS desde Windows. **“inactiva”** es normal, significa que no hay trabajos en cola (equivale a idle).



IMPLEMENTACIÓN DOCKPLOY, Nginex, POSTGRE SQL

Comprobar recursos de máquina virtual:

```
ubuntu@nutresa-server:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           3.8Gi       531Mi       2.6Gi       7.3Mi       1.0Gi       3.3Gi
Swap:             0B          0B          0B
ubuntu@nutresa-server:~$ df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        38G  2.4G   36G   7% /
ubuntu@nutresa-server:~$ nproc
2
ubuntu@nutresa-server:~$
```

Comprobar los puertos antes de instalar, como se declararon los puertos y no hubo ninguna respuesta, significa que están libres para su uso.

```
ubuntu@nutresa-server:~$ sudo ss -lntp | grep -E ':80 |:443 |:3000 '
```

Instalar Dokploy dentro de la máquina virtual declarando la ip utilizada en la mv:

```
ubuntu@nutresa-server:~$ sudo apt update
sudo apt install -y curl
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.16).
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
ubuntu@nutresa-server:~$
```

```
ubuntu@nutresa-server:~$ curl -sSL https://dokploy.com/install.sh | sudo ADVERTISE_ADDR=192.168.1.34 sh
Detecting latest stable version from GitHub...
Latest stable version detected: v0.31.0
Installing Dokploy version: v0.31.0
# Executing docker install script, commit: 8ad27bd3ac871feabe62f1d926c4067800c1732f
+ sh -c apt-get -qq update >/dev/null
+ sh -c DEBIAN_FRONTEND=noninteractive apt-get -y -qq install ca-certificates curl >/dev/null
+ sh -c install -m 0755 -d /etc/apt/keyrings
+ sh -c curl -fsSL "https://download.docker.com/linux/ubuntu/gpg" -o /etc/apt/keyrings/docker.asc
+ sh -c chmod a+r /etc/apt/keyrings/docker.asc
+ sh -c echo "deb [arch=amd64 signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu noble stable" > /etc/apt/
+ sh -c apt-get -qq update >/dev/null
INFO: Searching repository for VERSION '28.5.0'
INFO: apt-cache madison docker-ce | grep '28.5.0' | head -1 | awk '{$1=$1};1' | cut -d' ' -f 3
INFO: apt-cache madison docker-ce-cli | grep '28.5.0' | head -1 | awk '{$1=$1};1' | cut -d' ' -f 3
+ apt_flags=-y -qq
+ [ -n =5:28.5.0-1~ubuntu.24.04~noble ]
+ apt_flags=-y -qq --allow-downgrades
+ sh -c DEBIAN_FRONTEND=noninteractive apt-get -y -qq --allow-downgrades install docker-ce=5:28.5.0-1~ubuntu.24.04~noble docker-ce-cli=5
inerd.io docker-compose-plugin docker-ce-rootless-extras=5:28.5.0-1~ubuntu.24.04~noble docker-buildx-plugin docker-model-plugin >/dev/nu
Scanning processes...
Scanning linux images...
Using systemd to manage Docker service
+ sh -c systemctl enable --now docker.service 2>/dev/null
INFO: Docker daemon enabled and started

+ sh -c docker version
Client: Docker Engine - Community
```

```
de5cbc806753: Pull complete
Digest: sha256:31267173a15b4944e797a76ffd9c419707c8d8b32fe5b610f80cd0cfa05f3
Status: Downloaded newer image for traefik:v3.6.25
d19ae88104f34bb10824fb423a5cbf999e4cece6d592685e712ecf818fd9f9fd

Congratulations, Dokploy is installed!
Wait 15 seconds for the server to start
Please go to http://192.168.1.34:3000
If you are on the same local network, use http://172.31.38.240:3000

ubuntu@nutresa-server:~$
```

Comprobar que Dokploy quedó instalado:

```
ubuntu@nutresa-server:~$ sudo docker version
Client: Docker Engine - Community
 Version:           28.5.0
 API version:       1.51
 Go version:        go1.24.7
 Git commit:        887030f
 Built:             Thu Oct  2 14:54:28 2025
 OS/Arch:           linux/amd64
 Context:           default

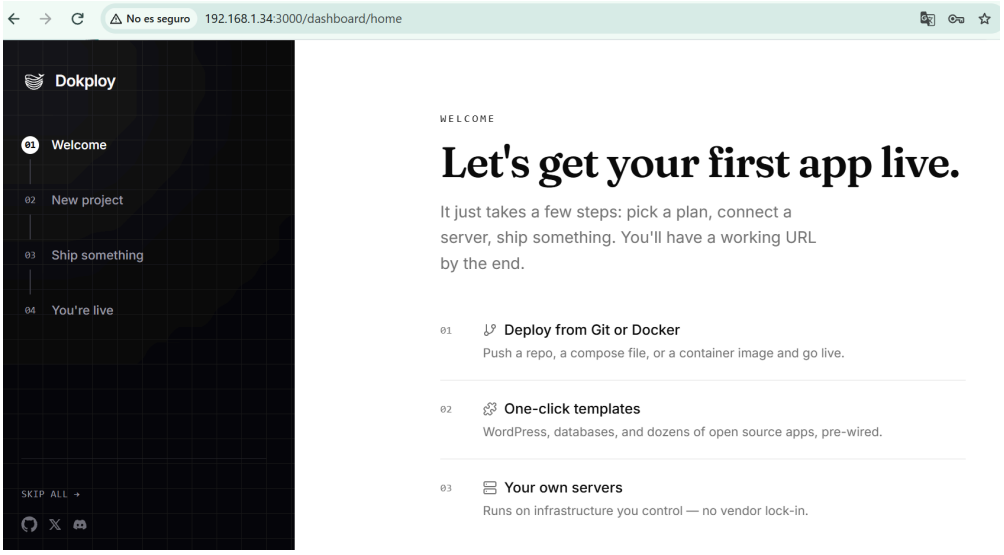
Server: Docker Engine - Community
 Engine:
  Version:          28.5.0
  API version:      1.51 (minimum version 1.24)
  Go version:       go1.24.7
  Git commit:       cd04830
  Built:            Thu Oct  2 14:54:28 2025
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          v2.4.2
  GitCommit:        b7a25261e029d68978bbf290df0a62fc819b3c51
 runc:
  Version:          1.5.2
  GitCommit:        v1.5.2-0-g29dd3dc2
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
ubuntu@nutresa-server:~$
```

```
ubuntu@nutresa-server:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-11-10 22:46:38 -05; 10min ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 14386 (dockerd)
      Tasks: 74
     Memory: 1.7G (peak: 2.5G)
        CPU: 2min 51.403s
     CGroup: /system.slice/docker.service
             ├─14386 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd
             ├─15492 /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 3000
             ├─15498 /usr/bin/docker-proxy -proto tcp -host-ip :: -host-port 3000 -con
             ├─15786 /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 80 -
             ├─15793 /usr/bin/docker-proxy -proto tcp -host-ip :: -host-port 80 -conta
             ├─15800 /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 443
             ├─15805 /usr/bin/docker-proxy -proto tcp -host-ip :: -host-port 443 -cont
             ├─15813 /usr/bin/docker-proxy -proto udp -host-ip 0.0.0.0 -host-port 443
             └─15822 /usr/bin/docker-proxy -proto udp -host-ip :: -host-port 443 -cont

Nov 10 22:46:48 nutresa-server dockerd[14386]: time="2026-11-10T22:46:48.843299566-05:
Nov 10 22:46:48 nutresa-server dockerd[14386]: time="2026-11-10T22:46:48.843318482-05:
Nov 10 22:46:48 nutresa-server dockerd[14386]: time="2026-11-10T22:46:48.843337337-05:
Nov 10 22:46:49 nutresa-server dockerd[14386]: time="2026-11-10T22:46:49.863834043-05:
Nov 10 22:47:20 nutresa-server dockerd[14386]: time="2026-11-10T22:47:20.071250168-05:
Nov 10 22:50:27 nutresa-server dockerd[14386]: time="2026-11-10T22:50:27.217758284-05:
Nov 10 22:51:44 nutresa-server dockerd[14386]: time="2026-11-10T22:51:44.358051927-05:
Nov 10 22:51:44 nutresa-server dockerd[14386]: time="2026-11-10T22:51:44.358249047-05:
Nov 10 22:56:44 nutresa-server dockerd[14386]: time="2026-11-10T22:56:44.358364177-05:
Nov 10 22:56:44 nutresa-server dockerd[14386]: time="2026-11-10T22:56:44.358442484-05:
```

```
ubuntu@nutresa-server:~$ sudo docker node ls
ID                            HOSTNAME         STATUS    AVAILABILITY   MANAGER STATUS   ENGINE VERSION
krnhe5ffardsjbn3gczhq85bv *   nutresa-server   Ready                    Leader           28.5.0
```

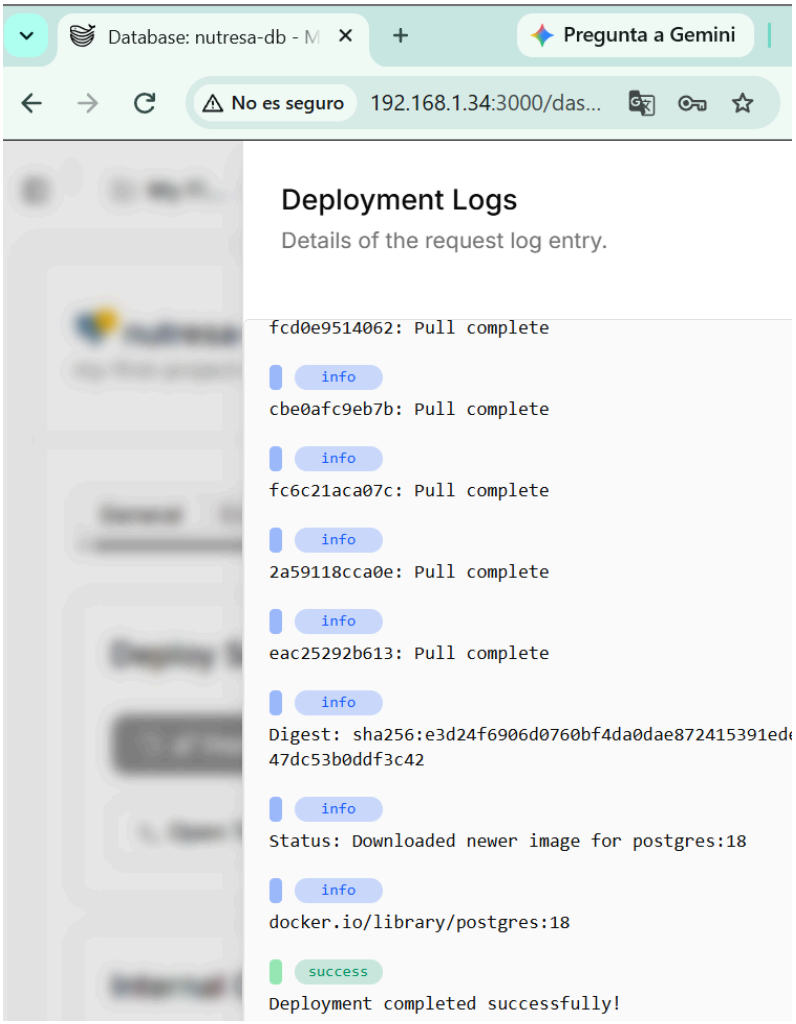
Abrir el panel de Dokploy: Funciona

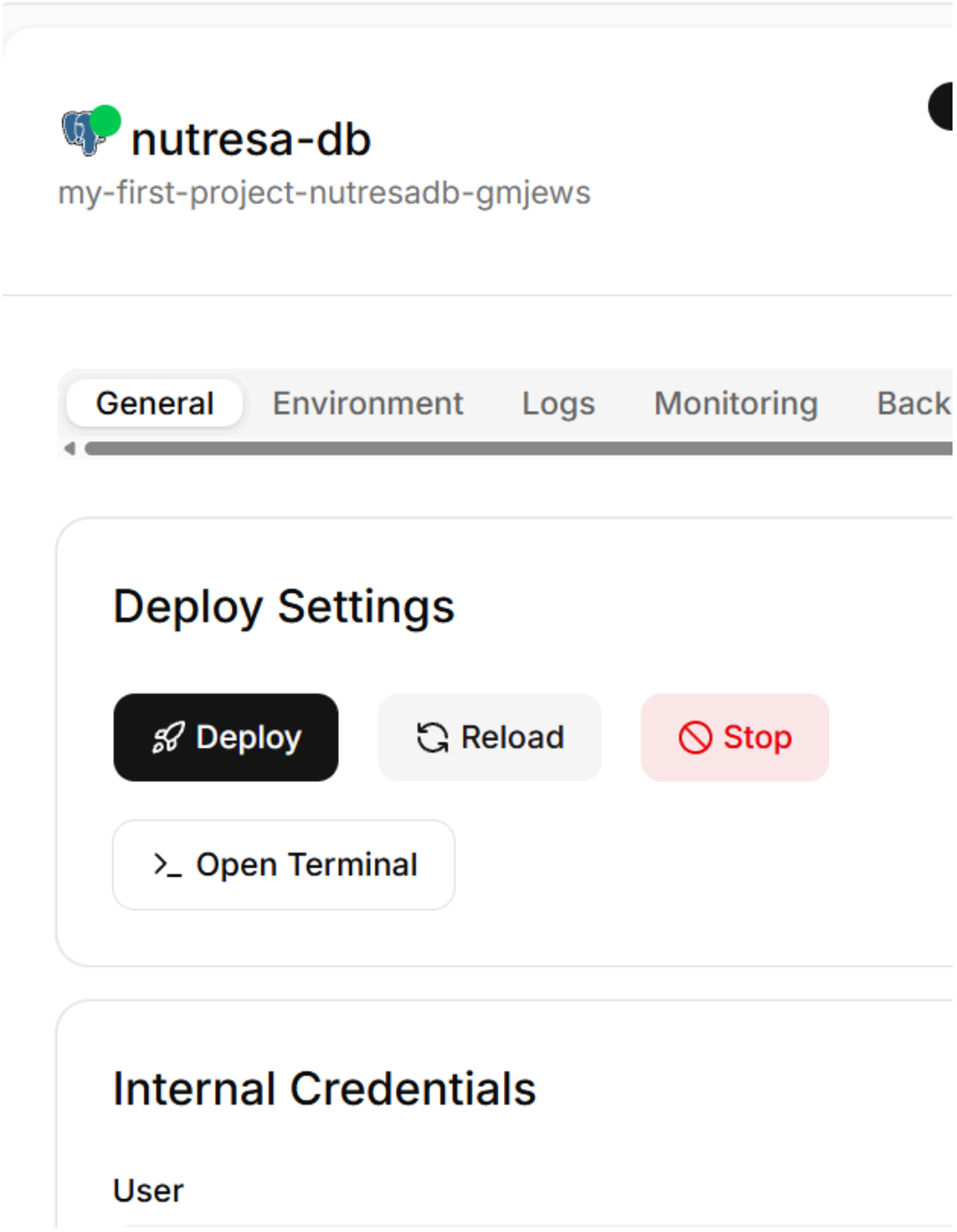


Restringir el acceso al panel:

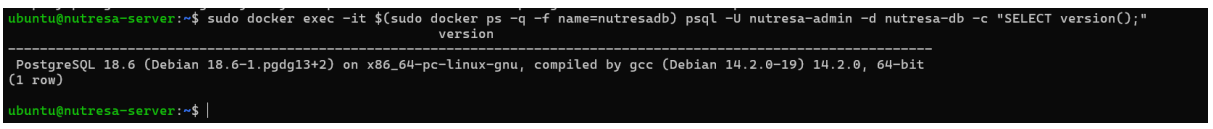
```
ubuntu@nutresa-server:~$ sudo docker node ls
ID                            HOSTNAME         STATUS    AVAILABILITY   MANAGER STATUS
krnhe5ffardsjbn3gczhq85bv *   nutresa-server   Ready     Active         Leader
ubuntu@nutresa-server:~$ sudo ufw allow from 192.168.1.31 to any port 3000 proto tcp
Rules updated
ubuntu@nutresa-server:~$ sudo ufw allow from 192.168.1.15 to any port 3000 proto tcp
Rules updated
ubuntu@nutresa-server:~$ sudo ufw allow from 192.168.1.0/24 to any port 80 proto tcp
sudo ufw allow from 192.168.1.0/24 to any port 443 proto tcp
Rules updated
Rules updated
ubuntu@nutresa-server:~$
```

Postgres Crear PostgreSQL desde Dokploy:





Comprobar PostgreSQL. En la guía debe aparecer 16, pero se hizo con la última actualización, por eso devuelve 18:



```
ubuntu@nutresa-server:~$ sudo docker exec -it $(sudo docker ps -q -f name=nutresadb) psql -U nutresa-admin -d nutresa-db
psql (18.6 (Debian 18.6-1.pgdg13+2))
Type "help" for help.

nutresa-db=# CREATE TABLE prueba_nutresa (
    id SERIAL PRIMARY KEY,
    mensaje VARCHAR(200)
);

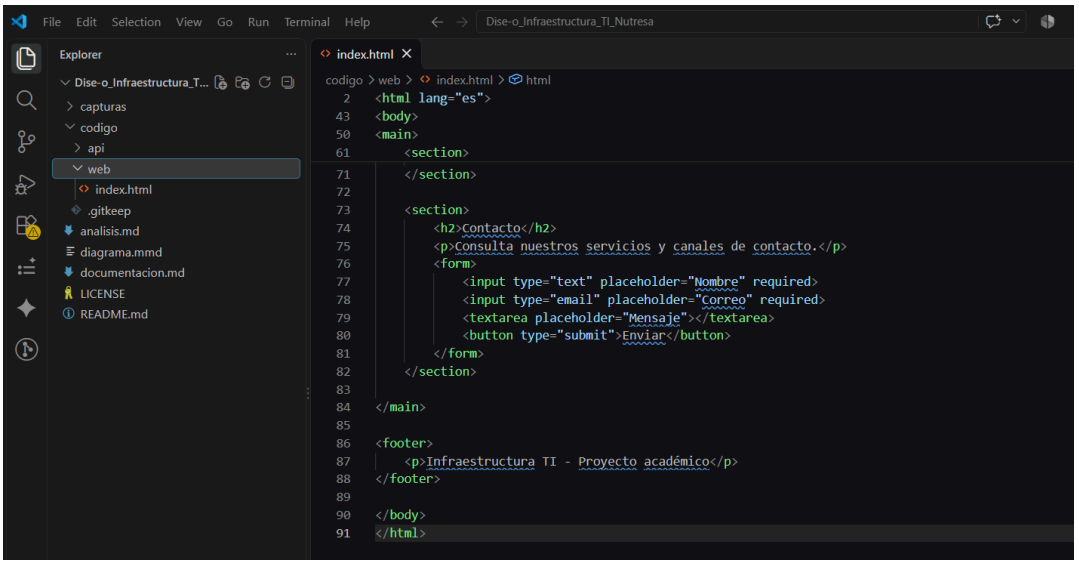
INSERT INTO prueba_nutresa (mensaje)
VALUES ('Base de datos Nutresa funcionando');

SELECT * FROM prueba_nutresa;
CREATE TABLE
INSERT 0 1
 id |              mensaje
----+-----------------------------------
  1 | Base de datos Nutresa funcionando
(1 row)

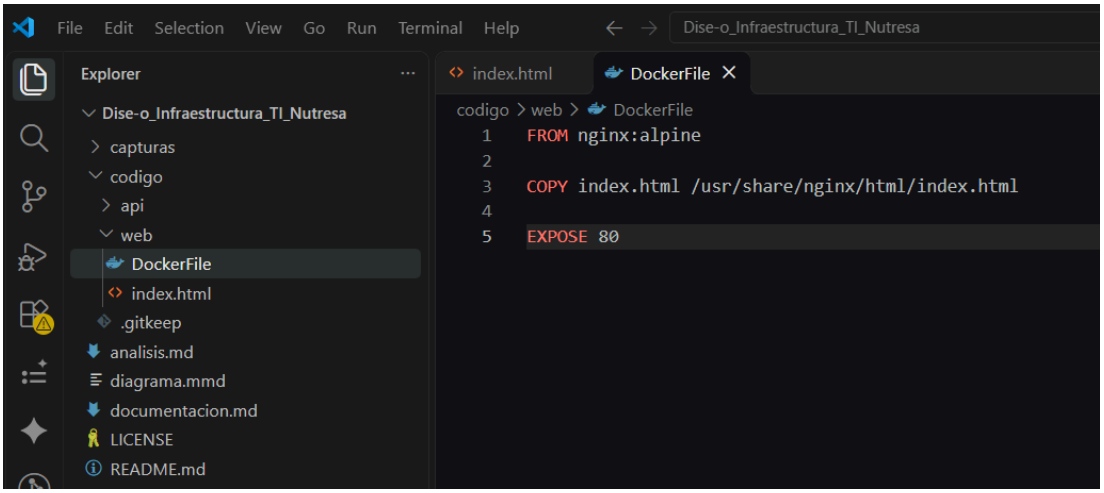
nutresa-db=# \q
ubuntu@nutresa-server:~$
```

WEB Y BACKEND API

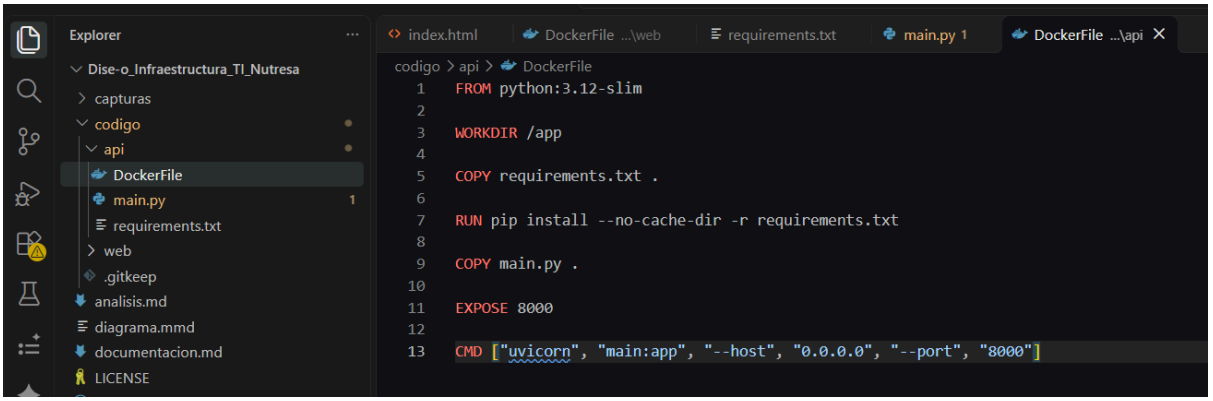
Crear la página web institucional:



Crear el Dockerfile de Nginx, Este archivo indica que la página se ejecutará dentro de un contenedor Nginx. :



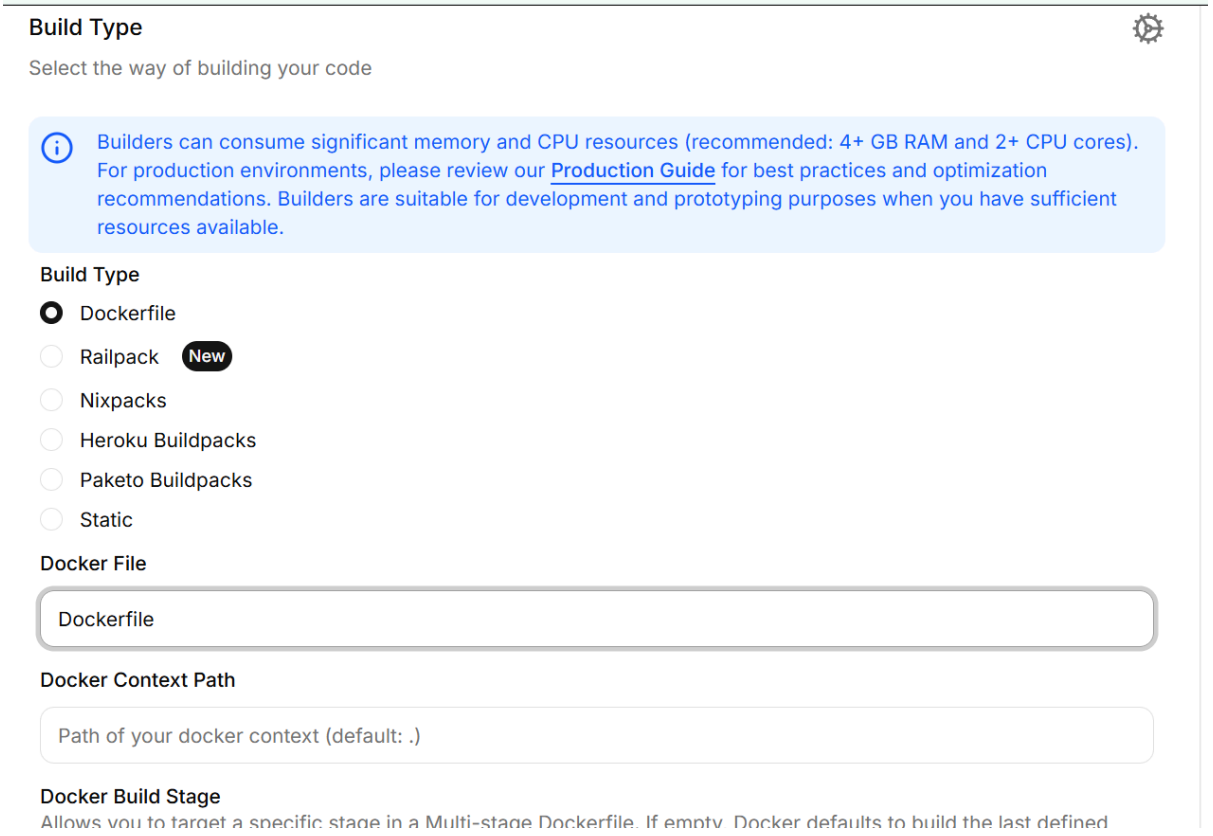
Crear el Backend API:



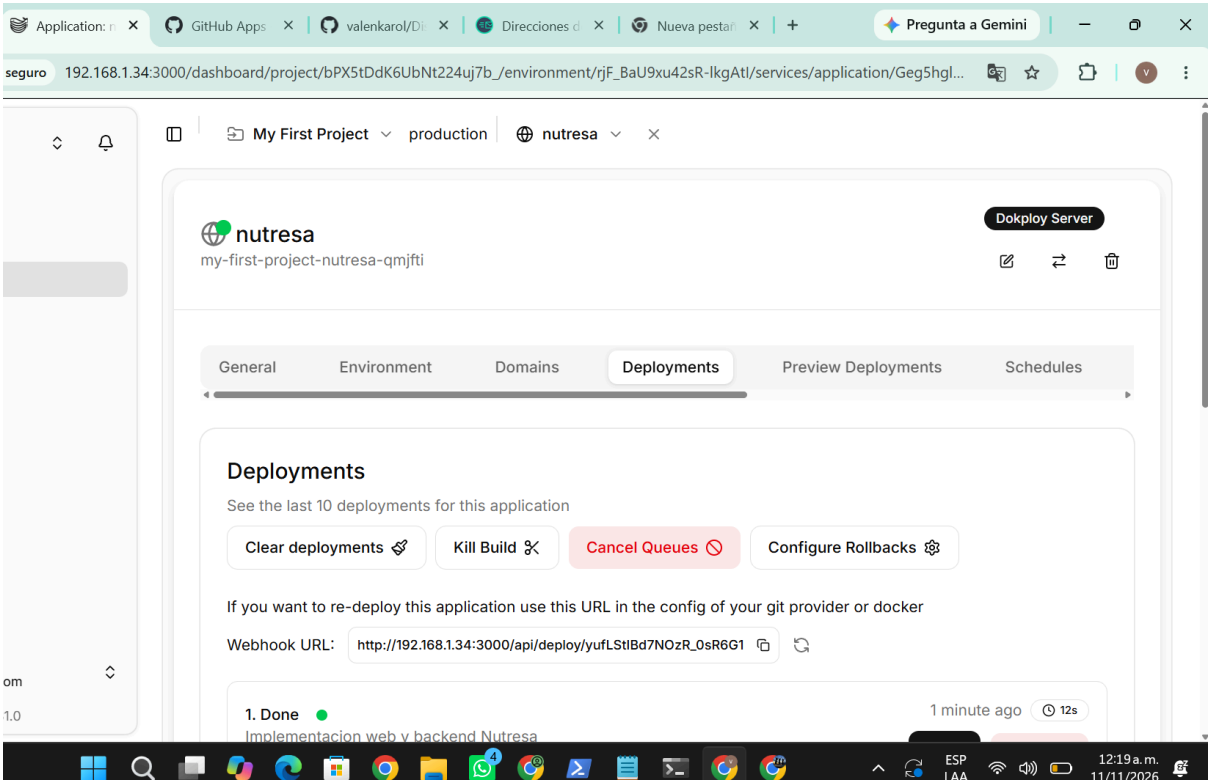
Esta API permitirá:

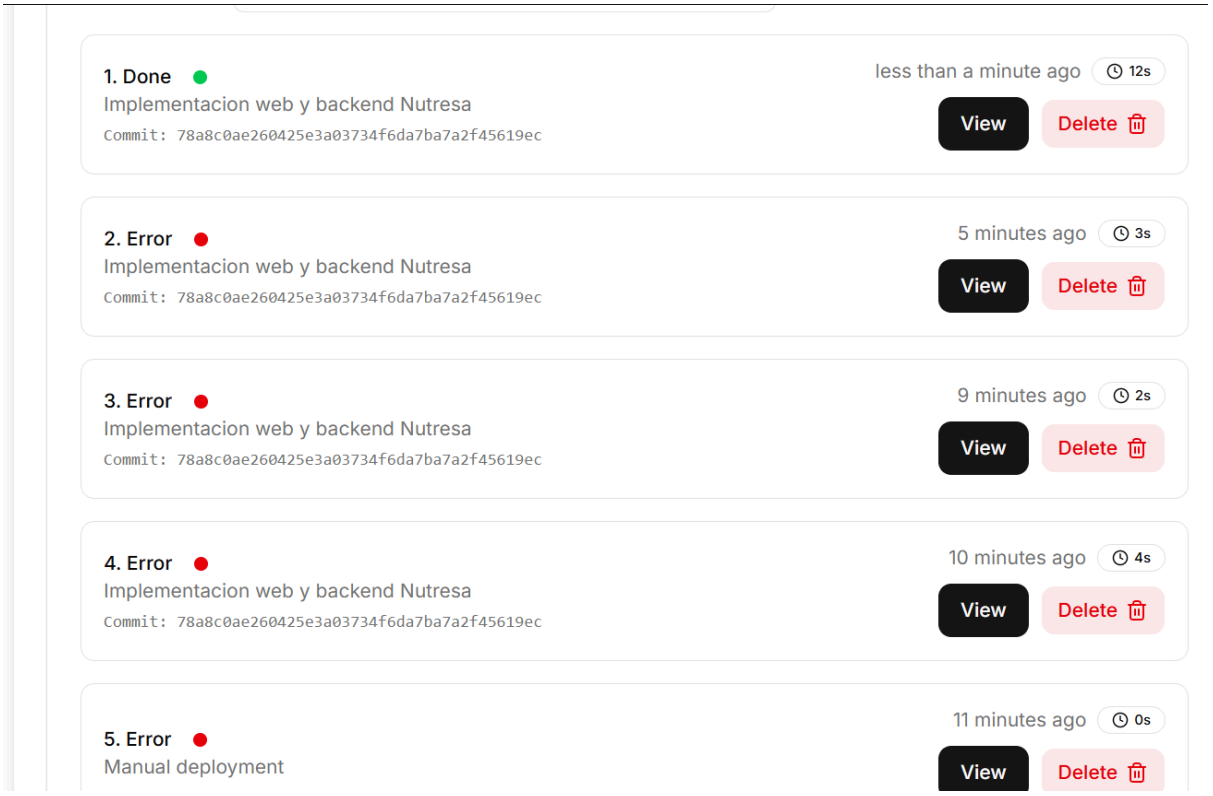
- Consultar su estado.
- Consultar productos.
- Registrar productos en PostgreSQL.

Para el despliegue de Nginx se conecto el dokploy con el repositorio:



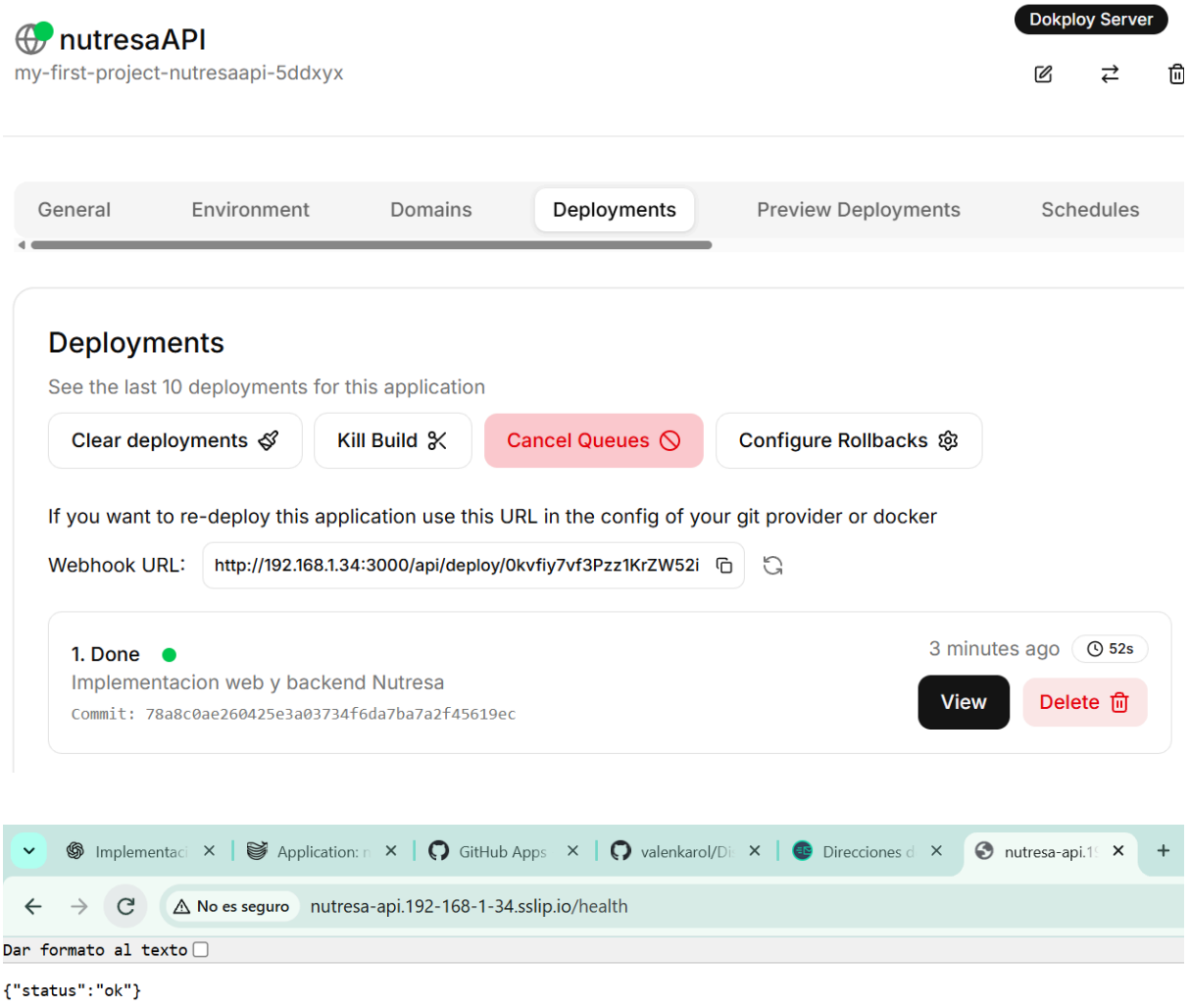
Antes de que se realizará el correcto despliegue se tuvo problemas con el nombre de uno de los archivos, luego se pudo solucionar y corrió correctamente.





Backend Api

Se empieza repitiendo los mismos pasos de la creación de la aplicación anterior, pero esta vez se llama al código de api.


nutresaAPI
my-first-project-nutresaapi-5ddxyx
Dokploy Server
General
Environment
Domains
Deployments
Preview Deployments
Schedules
Deployments
See the last 10 deployments for this application
Clear deployments
Kill Build
Cancel Queues
Configure Rollbacks
If you want to re-deploy this application use this URL in the config of your git provider or docker
Webhook URL:
http://192.168.1.34:3000/api/deploy/0kvfiy7vf3Pzz1KrZW52i
1. Done
Implementacion web y backend Nutresa
Commit: 78a8c0ae260425e3a03734f6da7ba7a2f45619ec
3 minutes ago
52s
View
Delete

Implementaci
Application: r
GitHub Apps
valenkarol/Di
Direcciones d
nutresa-api.1
No es seguro
nutresa-api.192-168-1-34.sslip.io/health
Dar formato al texto

{"status":"ok"}